Unity in Diversity?
How Intergroup Contact Can Foster Nation Building

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Diversity and Nation Building *Historically*

Uniting diverse groups is a founding principle of many nation-states

- **nation building** as promotion of shared national identity
  $\Longrightarrow$ weaken ethnic attachment, ameliorate intergroup divisions

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The creation of a nation—a people unified by ties of common language, common outlook, and common political participation, a people enthusiastically severing its outworn ties to local traditions and loyalties and achieving kesadaran, *consciousness of the nation . . .*

For some leaders the first task was the destruction of ethnic barriers and the creation in society at large of the sort of all-Indonesian culture which already existed inside the nationalist movement.

Feith, 1967

Diversity and Nation Building *Today*

Putnam (2007): *"(t)he most certain prediction that we can make about almost any modern society is that it will be more diverse a generation from now than it is today. . . the central challenge for modern, diversifying societies is to create a new, broader sense of we."*

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  - ▷ intergroup ties may form + Δ preferences over longer-run thru contact (Allport, 1954)
- Empirical **identification** of these effects of diversity **is difficult**:
  - ▷ local diversity often dissipates via tipping + segregation (Schelling, 1971)
  - ▷ long-run diversity confounded by geography and endogenous sorting (Michalopoulos, 2012)

How Does Intergroup Contact Affect Nation Building? Evidence from A Large-Scale Policy Experiment

- **Indonesia**: ethnolinguistically diverse, expansive archipelago
  - but, ethnic groups relatively isolated from each other historically
  - perennial challenges of regional separatism and nat'l disintegration
  - longstanding division: core Inner Island vs. periphery Outer Island



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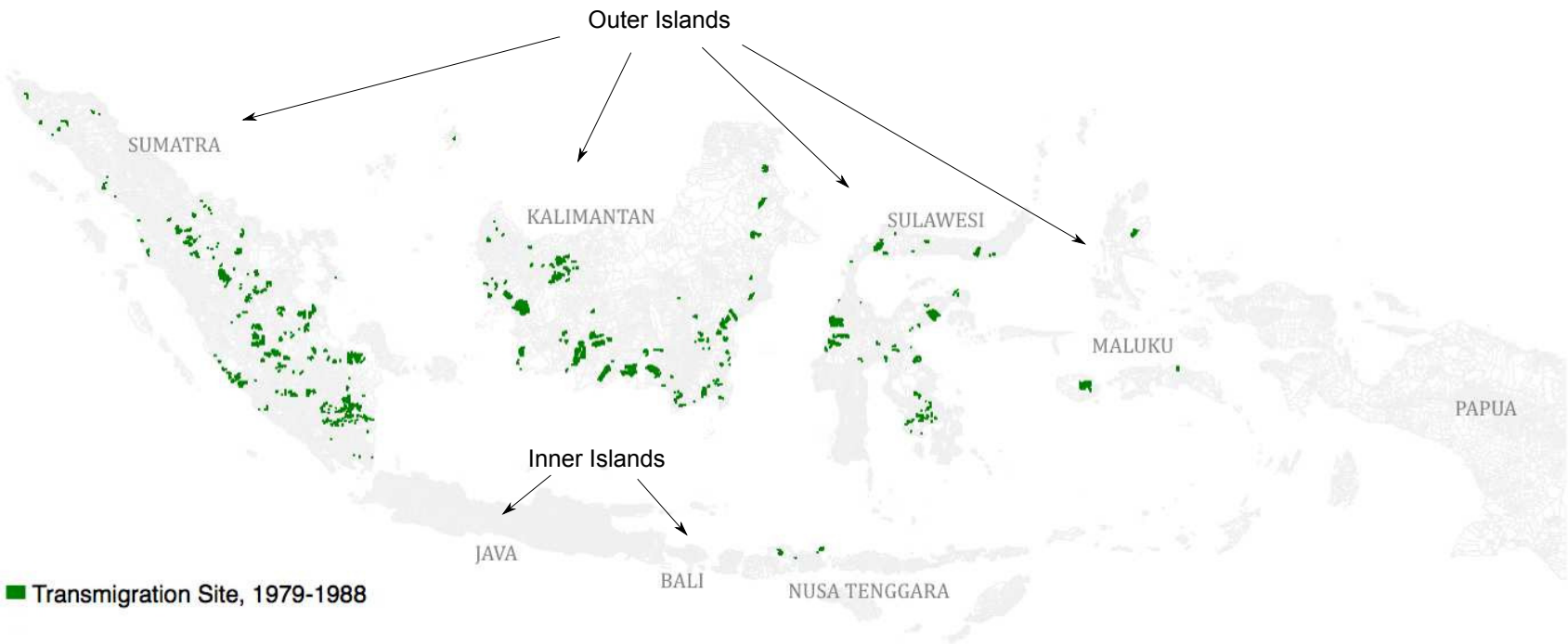
- **Population resettlement**: **as part of nation building policy**

$\Longrightarrow$ **persistent, plausibly exogenous long-run variation in local diversity**

A Natural Policy Experiment in Ethnic Mixing

Transmigration: voluntary rural-to-rural resettlement, 1979–1988

- ▷ 2 mn. migrants from Java/Bali placed in >**900 new villages**
- ▷ each community contained a **mix of Inner and Outer Islanders**
- ▷ goals of population redistribution + integration, but very controversial

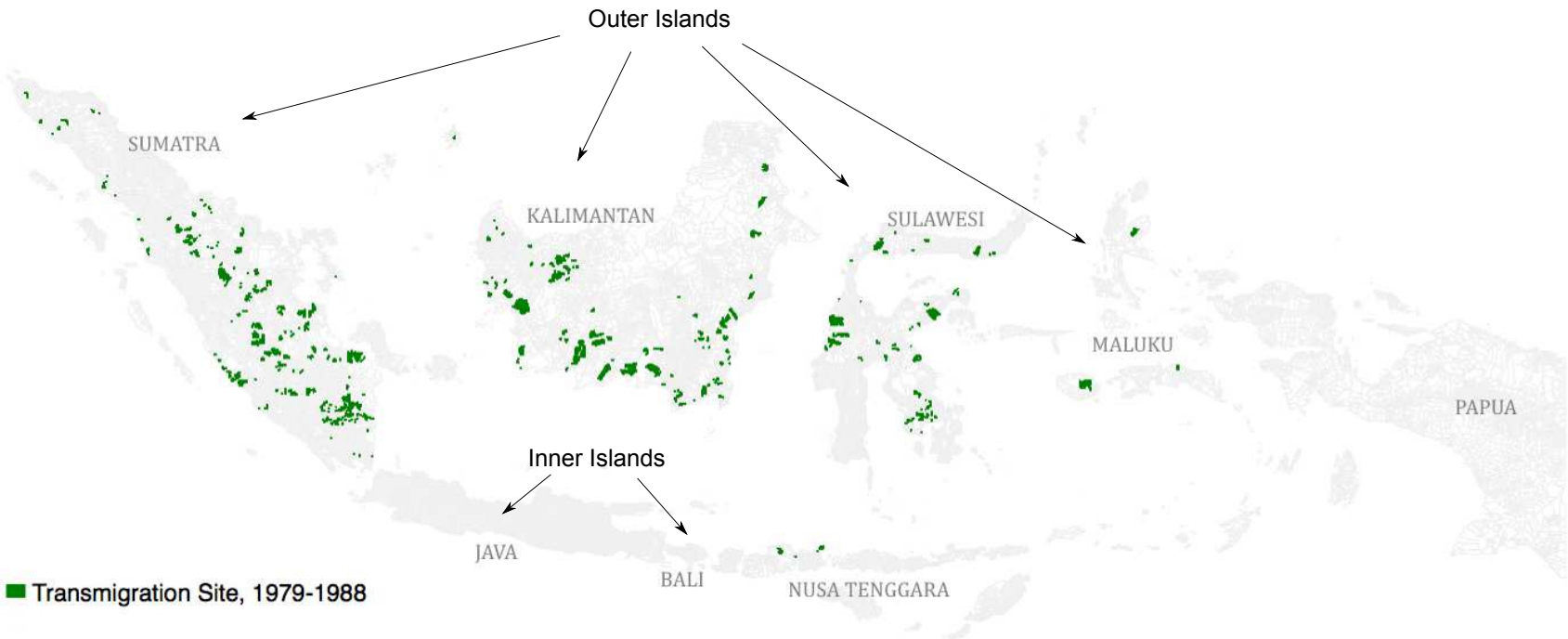


"[Transmigration] has been seen by national leaders as a tool for national integration . . . as a means of promoting cultural contact and building national unity." World Bank, 1988

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- Program: **initial, plausibly exogenous variation in diversity**
  ...persist thru migration frictions and property rights tying to land
  $\Longrightarrow$ **identify Δ incentives based on diversity, relative group sizes**

Preview of Results

Outcomes ∼12–25 Years Later

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(ii) cultural distance and integration vs. assimilation

(iii) interethnic economic competition

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3. **Policy discontinuity** $\Longrightarrow$ **ethnic mixing** $\Longrightarrow$ **nation building**

Contributions and Related Literature

1. Nation building (policy) amidst diversity and migration

(Alesina/Reich, 2015; Bandiera et al, 2016; Blouin/Mukand, 2016; Clots-Fig./Masella, 2013; Fouka, 2016; Laitin/Ramachandran, 2016; Miguel, 2004; Okunogbe, 2015)

2. Intergenerational process of cultural change in diverse societies

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Our key contributions to this growing literature:

- opportunity to identify convergence towards new form of shared identity, distinct from minority assimilation or conformity
  $\Longrightarrow$ national language as a revealed preference measure of identity
- conditions that facilitate integration + foster national identity
- nonlinear effects of diversity on integration$\rightarrow$cultural transmission
- policy-induced, long-run variation in diversity w/ limited sorting

Roadmap

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Data: Diversity and Nation Building Outcomes

Diversity, Socialization, and Identity Formation

Did the Transmigration Program Foster Nation Building?

Discussion

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Diversity and the Problem of Unity

An "Imagined Community" Amid National Diversity but Local Homogeneity

- Indonesia: **ethnolinguistically diverse** island nation
  - $>$ 1,000 ethnicities, 700 languages but living in homogeneous villages (median *village ELF* $= 0.05$, *national ELF* $= 0.7$)
  - 14 native **Inner-Island** groups: Java, Sunda, Bali, Madura largest
  - 900+ native **Outer-Island** groups: several large ones on each island (biggest groups: Minang, Bugis, Aceh, Batak, Banjar, Dayak, Toraja)
  - large diffs. in cultural norms across groups ▸ marriage, residence, inheritance
  - typical Inner–Outer linguistic differences $\approx$ German vs. French ▸

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- **Nation building** an important **concern for policymakers**
  - 1928 Youth Pledge: a pre-independence declaration of Indonesian unity
  - "Unity in diversity": national motto (also in E.U.), enshrined in coat of arms



Identity Building Through Language Policy

- *Bahasa Indonesia*, a key item in the 1928 declaration of unity
- root in lingua franca *Malay* **not** plurality Javanese ($\sim$ 40% of pop.)
- 1930s: roughly 5% of the population able to speak Indonesian
- today: $\sim$95% can speak; 18% use as primary language at home

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AsiaBarometer: individuals reporting primarily Indonesian use at home $\Longrightarrow$ 15% $\uparrow$ attachment to Indonesian rather than own ethnic identity

Resettlement as Part of Nation Building under Suharto

- **Transmigration: large-scale resettlement in late 1970s**
  - concerns about density: Java/Bali 66% of pop., 7% of land
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- Skeptics viewed program as vehicle for 'Javanization' of Outer Islands
  (Charras et al, 1993; Levang, 1995; Schiller & Ganang, 2002)
- Popular fears of violent conflict between Inner and Outer Islanders
  (lots of anecdotes + claims in Fearon & Laitin, 2011 re Papua)

Transmigration Program Implementation

- **Selecting Sites**
  - **new villages and farms** created on previously uncleared federal land
  - **site selection** based on geographic and agroclimatic features (**x**) (topography, soil quality, water access, weather, transport access)

    $\Longrightarrow$ **recommended development areas** (RDAs)

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- **Designing New Settlements**
  - **carrying capacity** based on land quality and quantity
  - *de jure*, 10–30% **slots for local Outer-Island natives** (APDDT)
    de facto, some settlements included as high as 50–80%
  - house + 2 ha farm plots **allocated by lottery**, ownership after 5-10 yrs
  - identical public institutions (schools, gov't office) in all settlements

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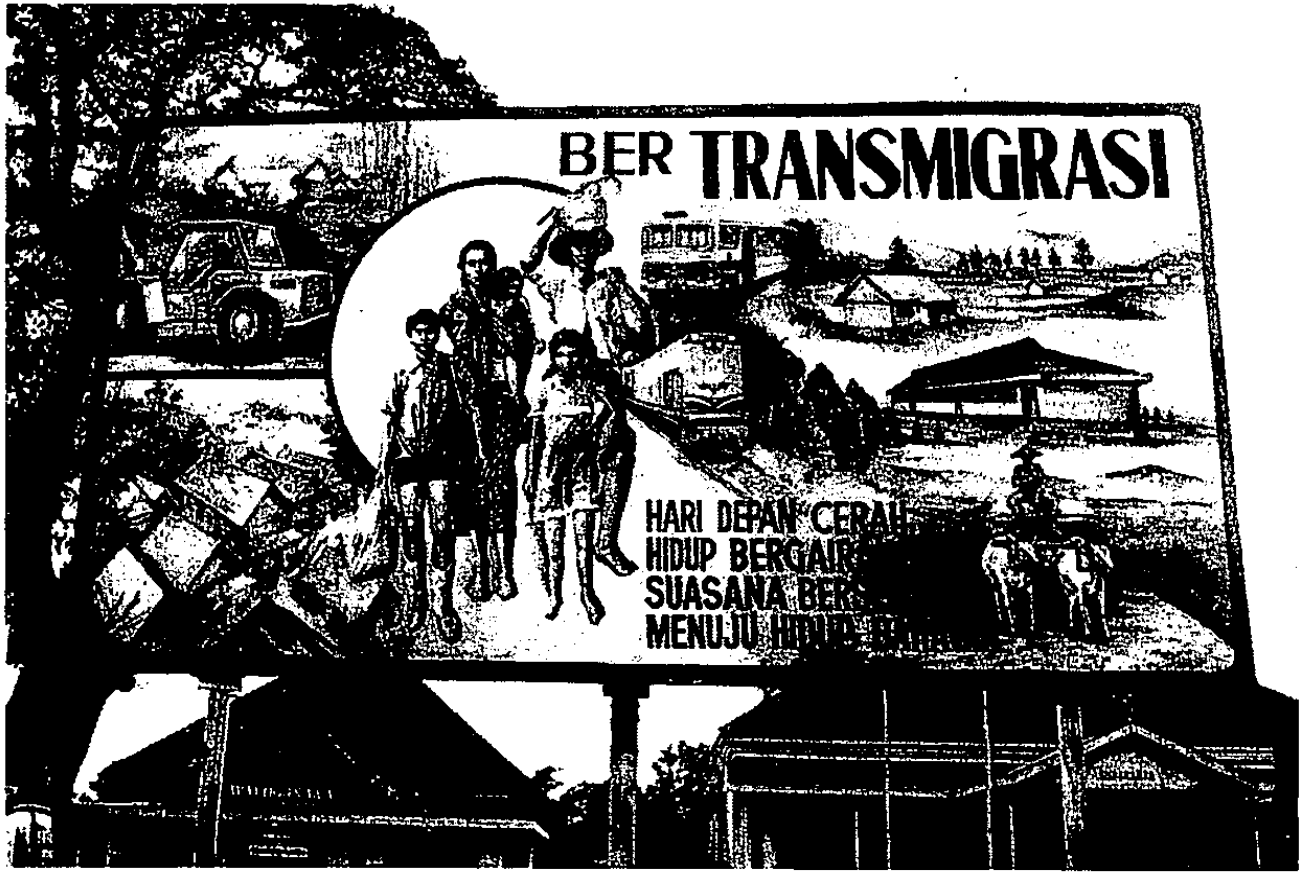
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Transmigrant Households

- Voluntary participation: married, farmers, household head age 20-40 (low schooling, similar to non-migrants from rural Java/Bali) ▸ table

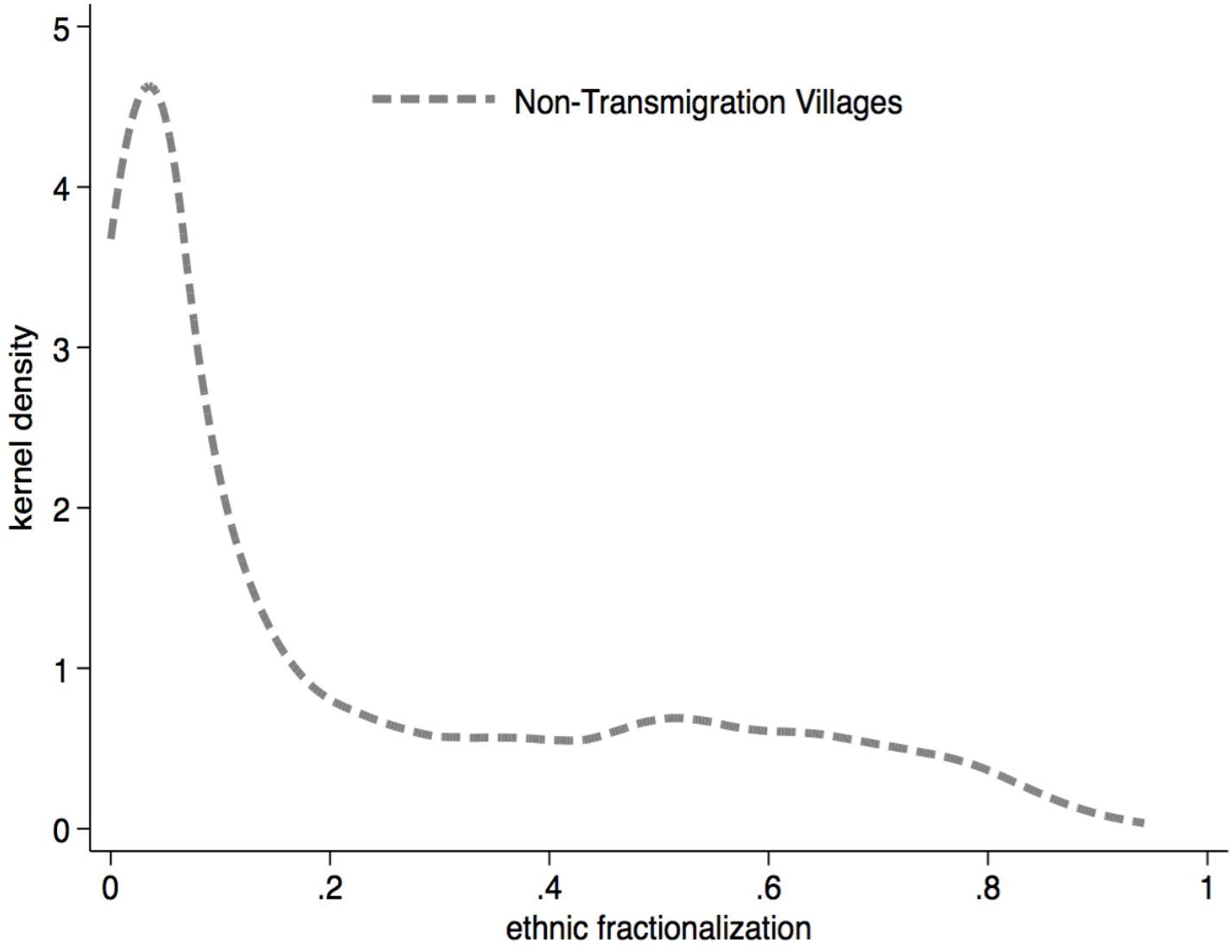
Advertising the Transmigration Program

Billboard in rural Java, early 1980s



"A bright and vigorous future, together we move towards a joyous life"

What Does the **Policy Experiment** Buy Us?
Persistent Continuum of Local Diversity



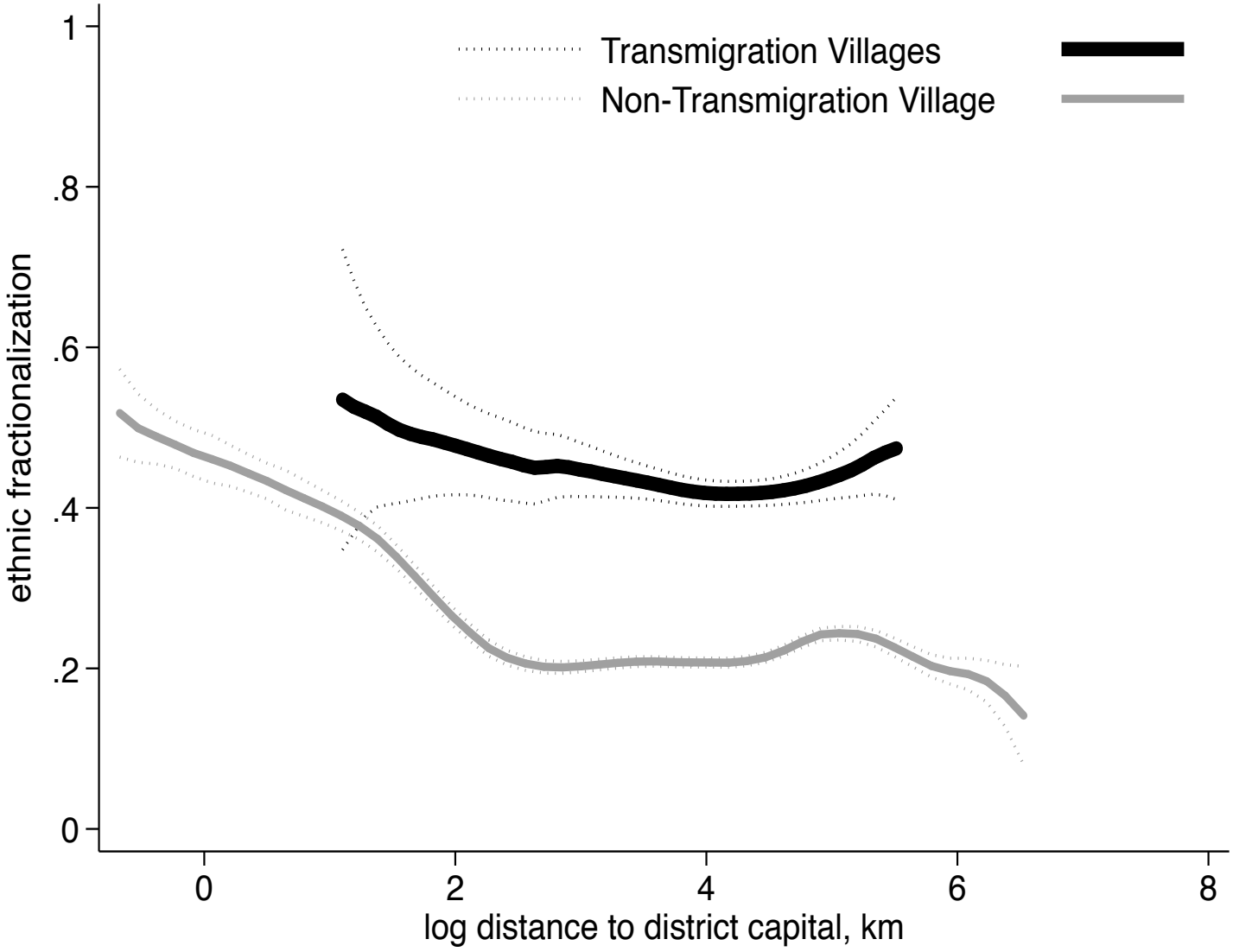
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What Does the **Natural Experiment** Buy Us?
Diversity More Exogenous w.r.t. Natural Advantages

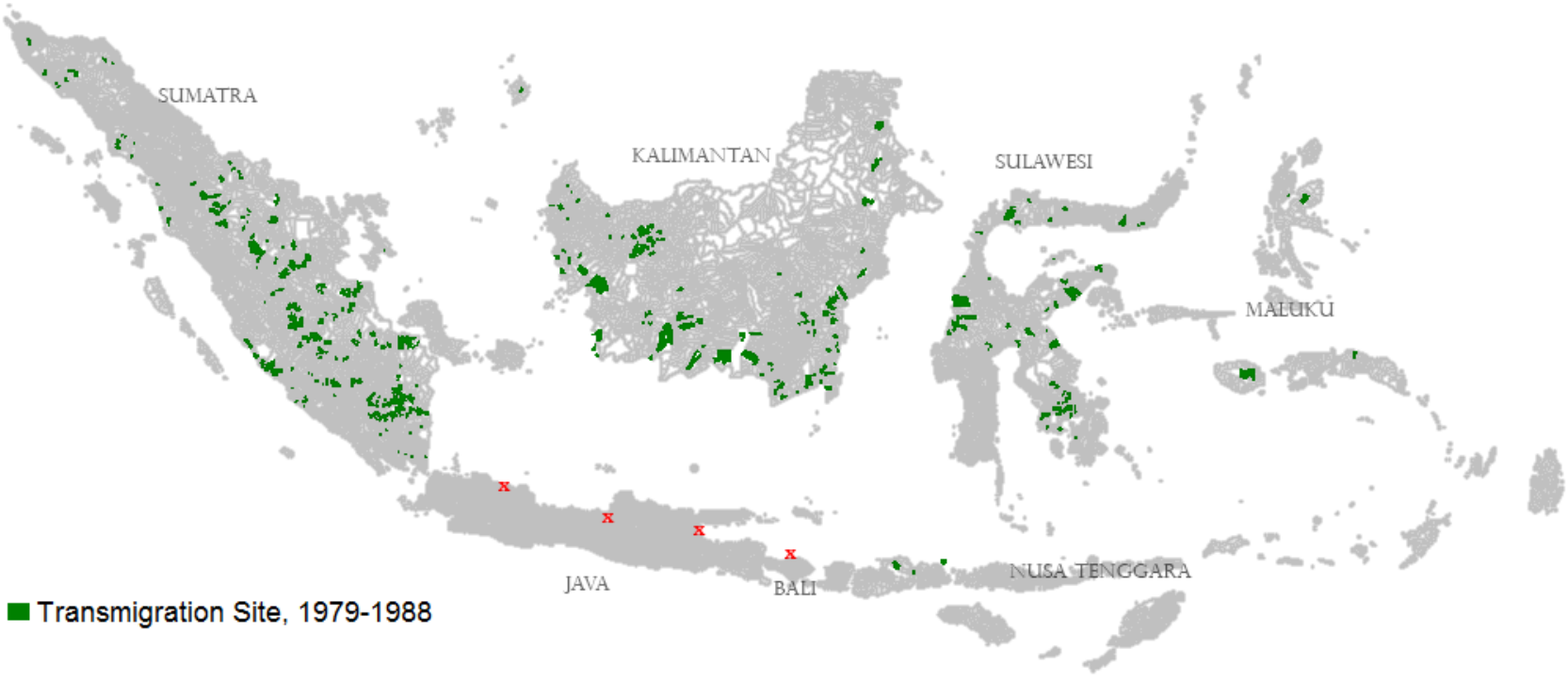


What Does the **Natural Experiment** Buy Us?
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Lack of a Systematic Assignment Mechanism

- Transmigrants sent from 4 transit camps (x) and could not choose destinations
  - knew very little pre-departure re destinations; 85% did not know local ethnic group (Kebschull, 1986 camp survey)



- *plan-as-you-proceed*: "we would just ship out groups of transmigrants as they showed up in transit camps" (arbitrary queuing with institutional constraints)

$\Longrightarrow$ **exogenous variation in ethnic mix of, distance b/t Inner and Outer Islanders**

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Measuring Ethnic Diversity and Segregation

- 2000 Population Census: reports *one ethnicity* for each individual
  - universal coverage; census block; place of birth and residence in 1995
  - identify **interethnic marriage status of household head** (10.7% exogamous/out-group marriages across country)

- **Inner-Island ethnic share**: diversity maximized at 50–50 split
  - $>$ 70% of variation in overall *ELF* explained by Inner-Island share (diversity w/in transmigrants, but local native pop. fairly homogenous)
  - $>$ 90% of variation due to 1st and 2nd gen. Inner-Island immigrants

- **Ethnic residential segregation** within-village: isolation index (Bell, 1951)

summary stats

Language Outcomes: Culture and Identity

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- 2006 Household Survey: ethnicity + **main language at home**
  - ▷ single household head responding
  - ▷ languages grouped into Indonesian, native Inner, native Outer

- 1995 Household Survey: language use at home + **mother tongue**
  - ▷ all household members responding; years of residence in village

  $\Longrightarrow$ **intergenerational transmission of (linguistic) identity**

National Language as Vehicle for Nation Building

Socialization $\rightarrow$ Intergenerational Transmission of Preferences

Using the *Indonesia Family Life Survey*, we relate individual outcomes in 2014 to their parents' choices from their former household in 1997:

$$y_{ij}^{14} = \alpha + \eta \text{Indonesian at home}_{ij}^{97} + \mathbf{x}_{ij}'\boldsymbol{\delta} + \theta_j + \varepsilon_{ij}$$

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	Dependent Variable as Adult in 2014:			
	Speaks Indonesian at Home	Changes Ethnicity from 1997	In Interethnic Marriage	Trust Other Ethnic Groups (z-score)
	(1)	**(2)**	**(3)**	**(4)**
Indonesian was Primary Language at Home as Child in 1997	0.156 (0.022)***	0.062 (0.019)***	0.053 (0.023)**	0.148 (0.054)***
Dependent Variable Mean	0.369	0.114	0.103	0.00
Age, Gender, Education Fixed Effects	Yes	Yes	Yes	Yes
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Childhildhood Indonesian use ∼ weaker ethnic attachment

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1. Diversity and Language Use at Home	diversity thresholds mechanisms
2. Diversity and Intergenerational Transmission	mother tongue horizontal vs. vertical transmission
3. Place-Based Impact: ATT	$\uparrow$ diversity $\uparrow$ national language use at home $\uparrow$ intermarriage + mechanisms

Ethnic Diversity and National Language Use at Home

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And, these effects may be:

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Extended Lazear (1999) model w/ nat'l language $\implies$ multiple equilibria

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Instrumental Variables Strategy

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- *diversity* = Inner-Island ethnic share *or* overall *ELF* in 2000
- Problem: *diversity* = *ex ante* assignment + *ex post* sorting

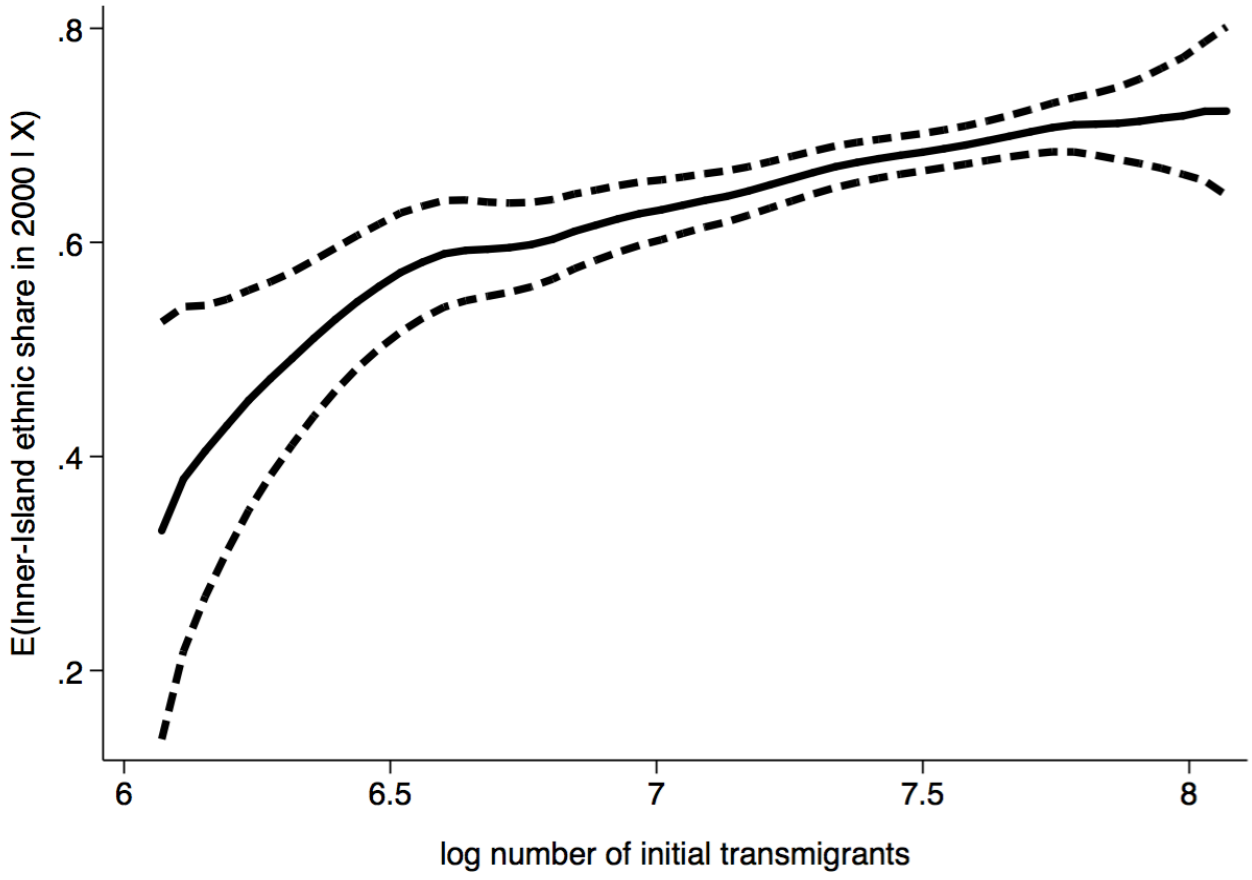
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- Problem: *diversity* = *ex ante* assignment + *ex post* sorting
- We propose **program-based instruments**
  - ▷ **# transmigrants assigned** 1979–88 $\Longrightarrow$ Inner-Island ethnic share
  - ▷ . . . + **fractionalization(Inner)** $\Longrightarrow$ fractionalization(overall)
- Conditioning on carrying capacity (**x**) isolates the implied local share
- **Intuition**: isolate portion of $diversity_j$ driven by *ex ante* assignment

Ethnic Diversity and National Language Use at Home
Instrument Strength

Diversity in 2000 is Strongly Predicted by Initial Transmigrants



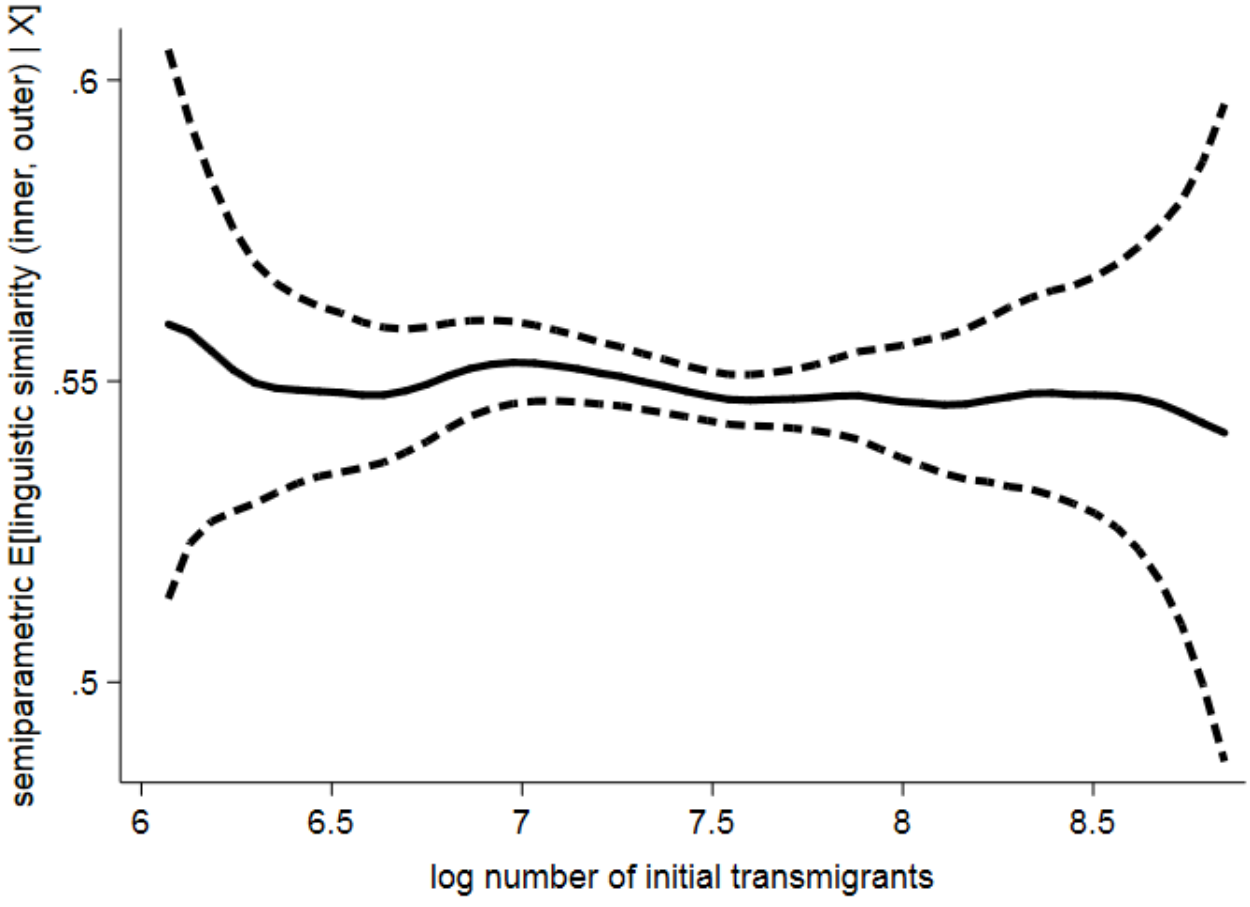
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Probing Instrument Validity

- Several results support **excludability**. Our instrument, the initial number of Inner-Island settlers, is uncorrelated with:

    1. linguistic distance b/t Inner and indigenous Outer-Island ethnicity
    2. ex post immigration (by group) between 1995 and 2000
    3. agroclimatic similarity of transmigrants (proxy for economic welfare)

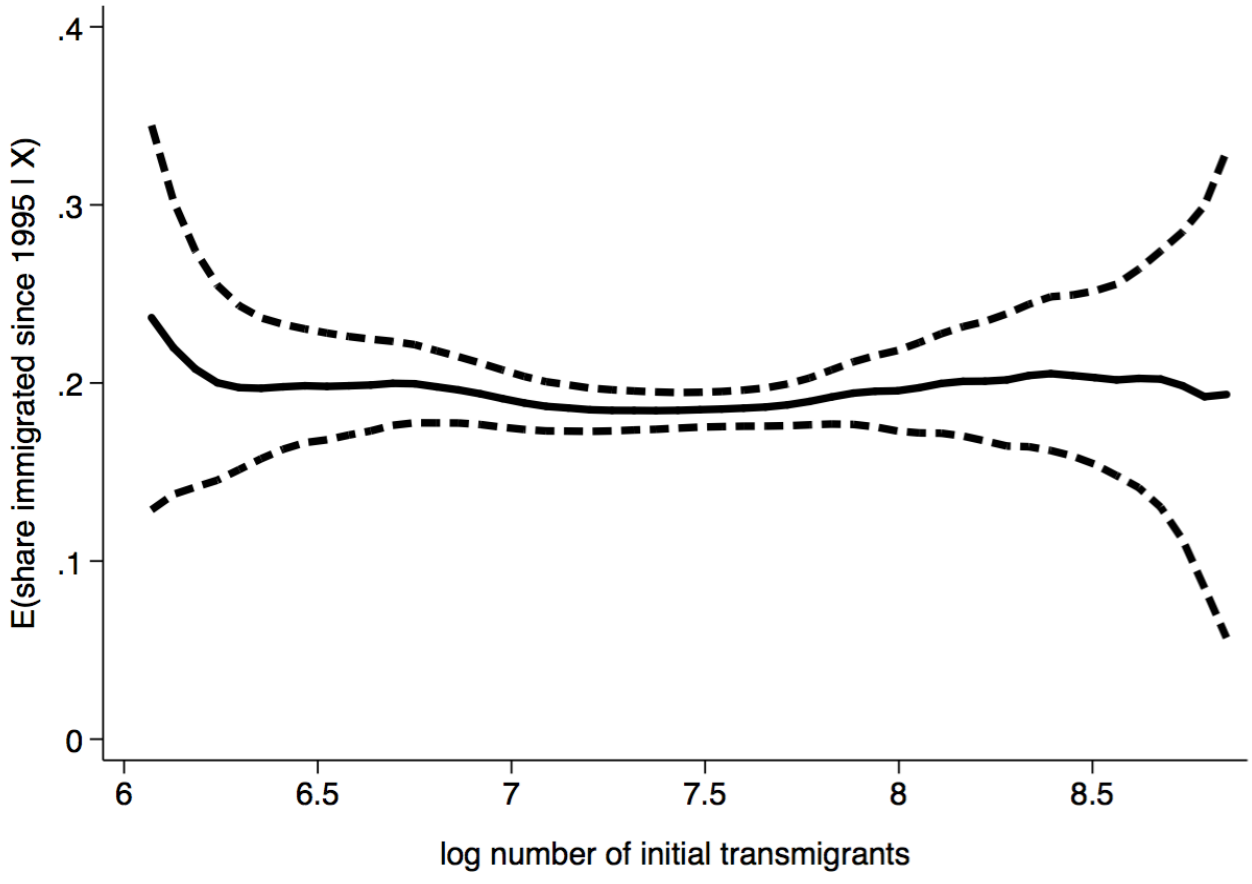
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1. Orthogonal to Linguistic Distance with Indigenous Ethnic Group



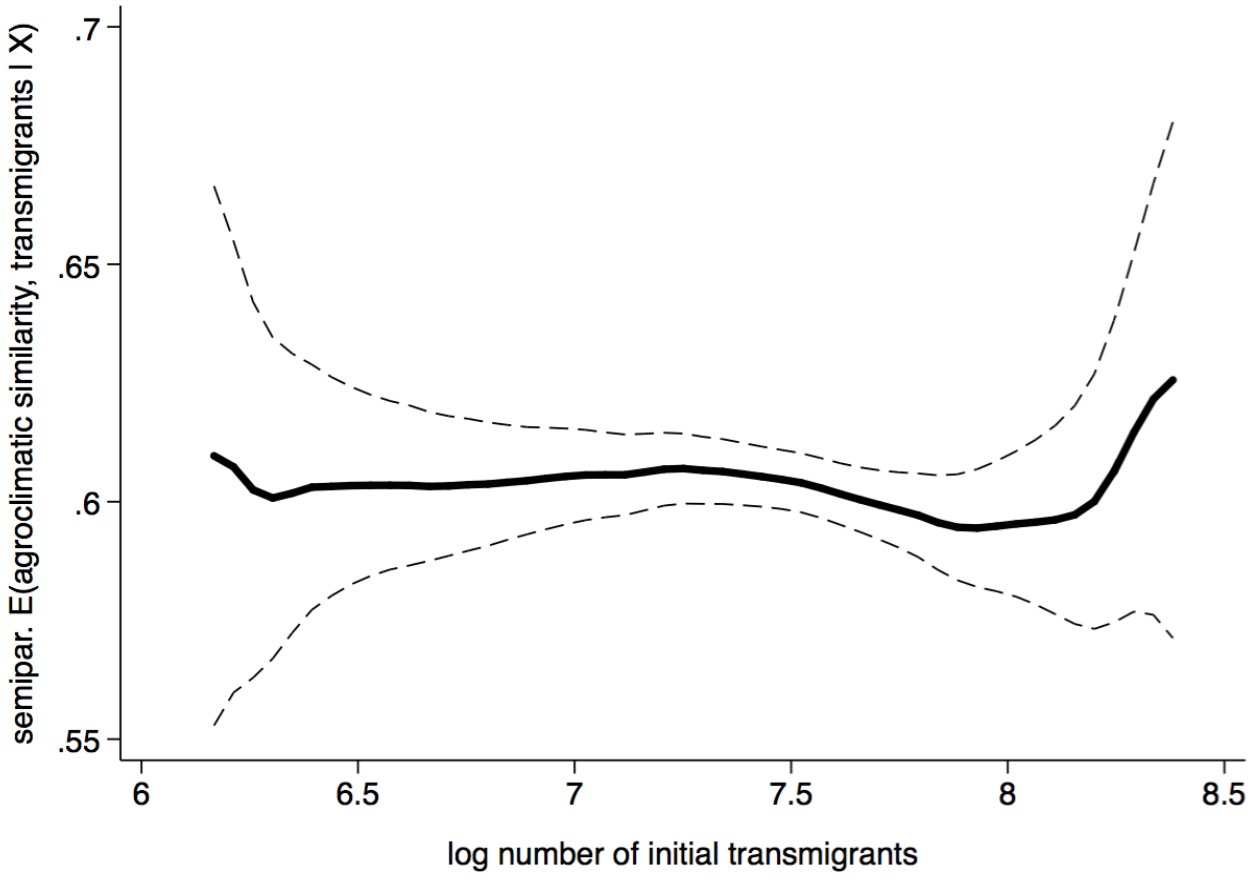
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2. Orthogonal to Ex Post Immigration Between 1995 and 2000



Ethnic Diversity and National Language Use at Home
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3. Orthogonal to Agroclimatic Similarity



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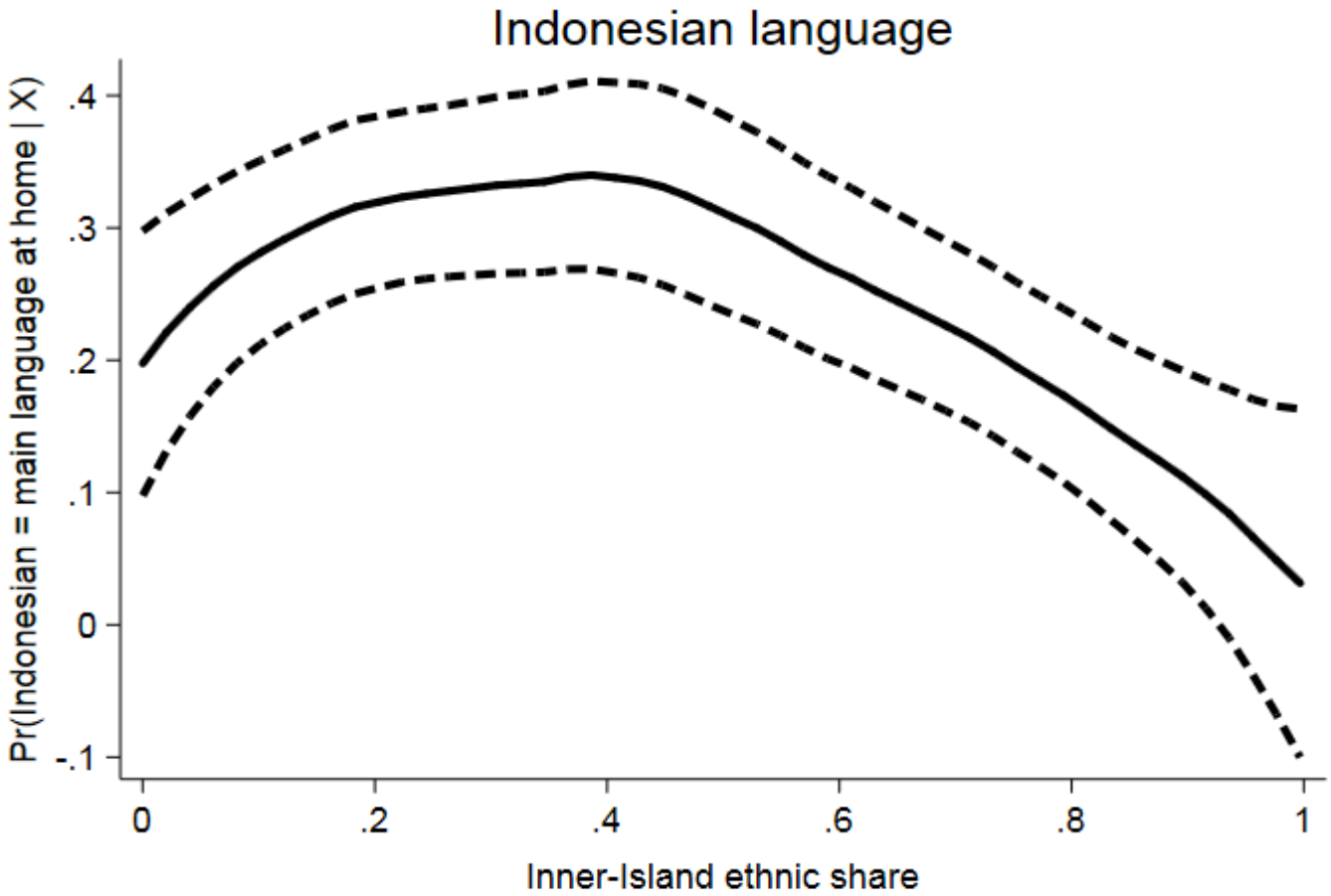
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  2. ex post immigration (by group) between 1995 and 2000
  3. agroclimatic similarity of transmigrants (proxy for economic welfare)
  4. other measures of diversity (e.g., birthplace or religious diversity)
  5. other measures of predetermined local political and economic development not explicitly used by the planners

Ethnic Diversity and National Language Use at Home
OLS Estimates of the Nonlinear Shape $g(\cdot)$

$$\textit{Indonesian}_{ij} = \alpha + g(\textit{Inner-Island ethnic share}_j) + \mathbf{x}'_{ij}\boldsymbol{\beta} + \varepsilon_{ij}$$

Ethnic Diversity and National Language Use at Home
OLS Estimates of the Nonlinear Shape $g(\cdot)$

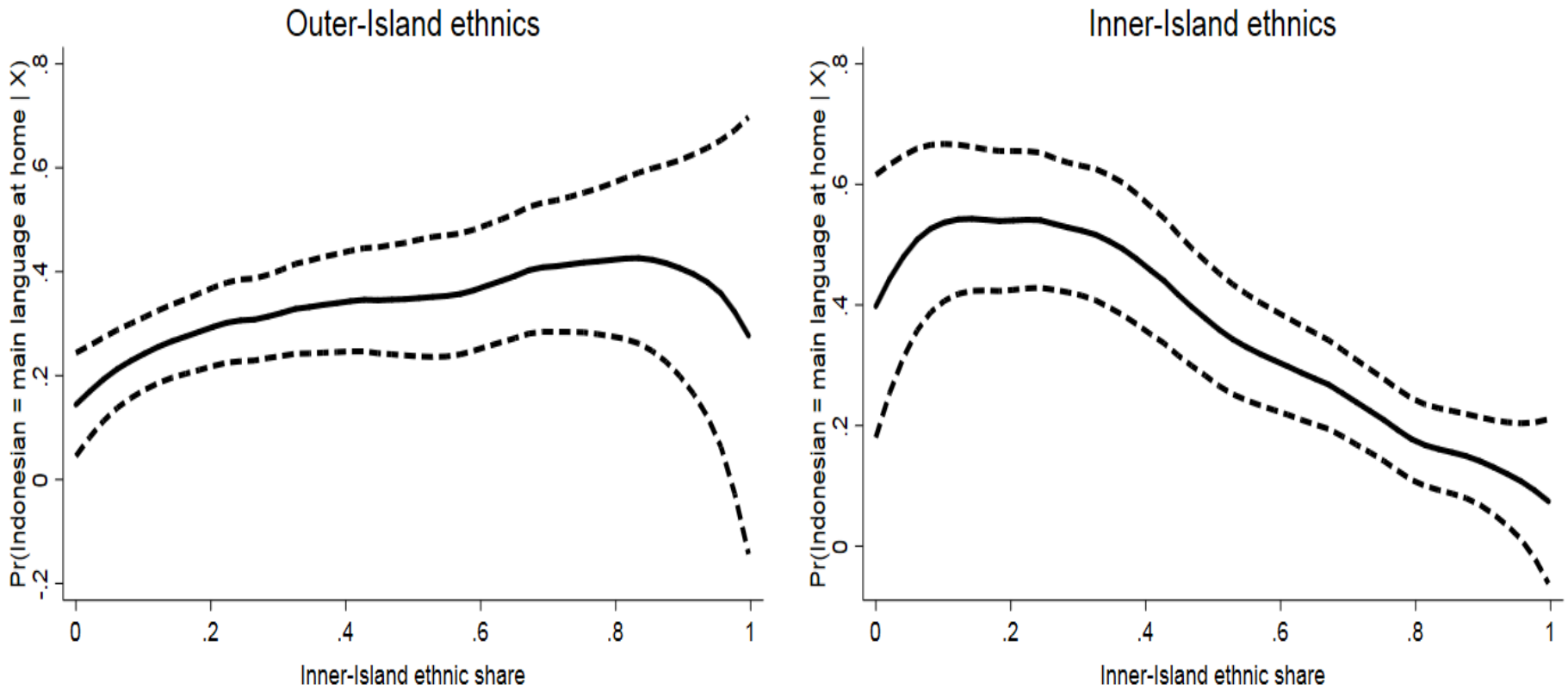
$$\textit{Indonesian}_{ij} = \alpha + g(\textit{Inner-Island ethnic share}_j) + \mathbf{x}'_{ij}\boldsymbol{\beta} + \varepsilon_{ij}$$



Notes: ▸ Robinson (1988) partially linear approach to estimating $g(\cdot)$. Based on *Susenas* data. Similar results using ▸ *Supas*.

Ethnic Diversity and National Language Use at Home
Understanding the Nonlinear Shape $g(\cdot)$

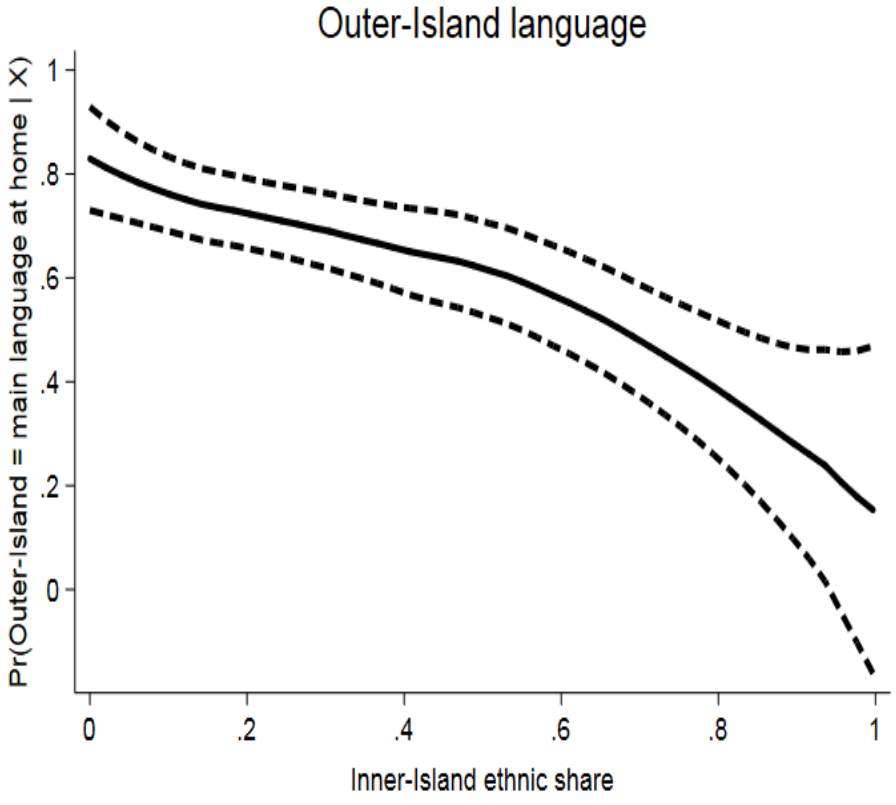
National Language Use at Home by Ethnic Group



Linear relationship for each group with similar slope.

Ethnic Diversity and National Language Use at Home
Understanding the Nonlinear Shape $g(\cdot)$

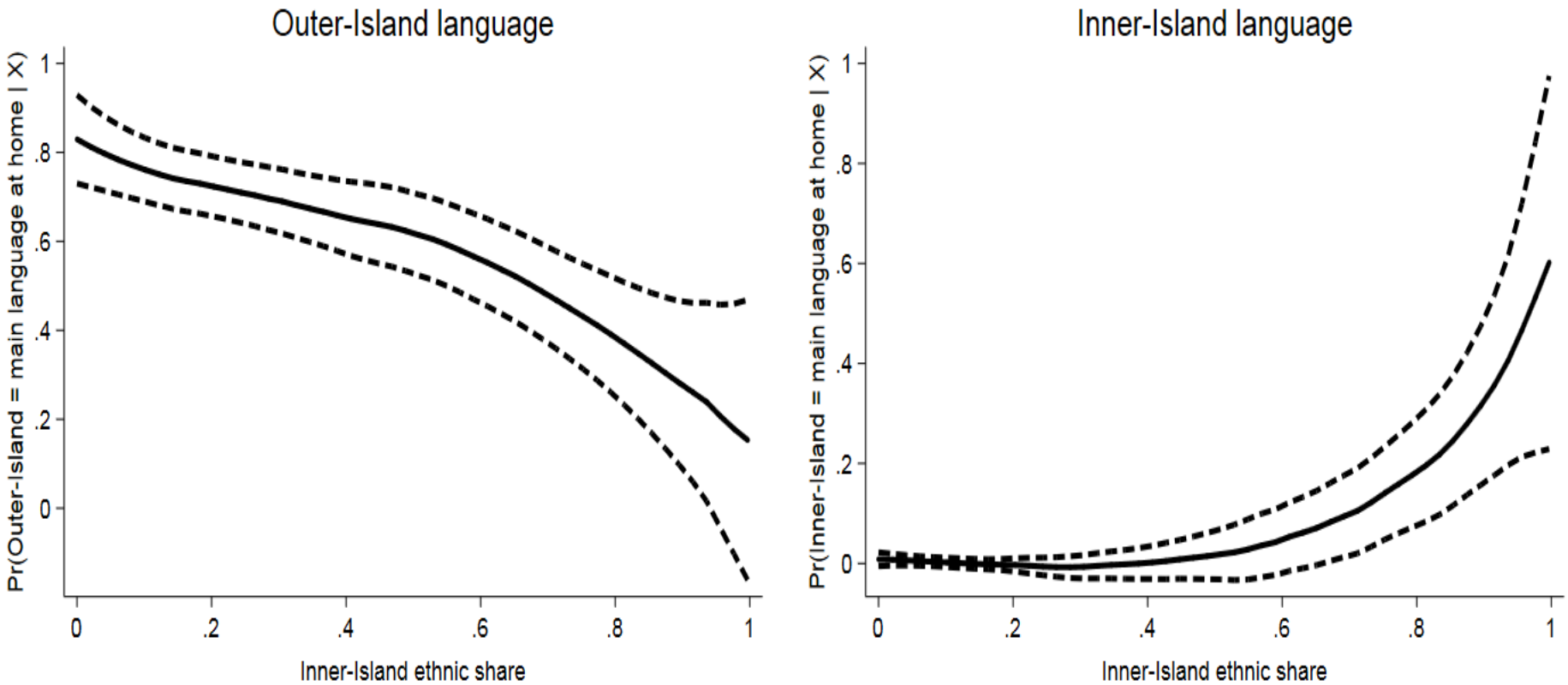
Language Use at Home by Outer-Island Ethnic Groups



LEFT: Linearity for own language, consistent w/ weakened ethnic attachment.

Ethnic Diversity and National Language Use at Home
Understanding the Nonlinear Shape $g(\cdot)$

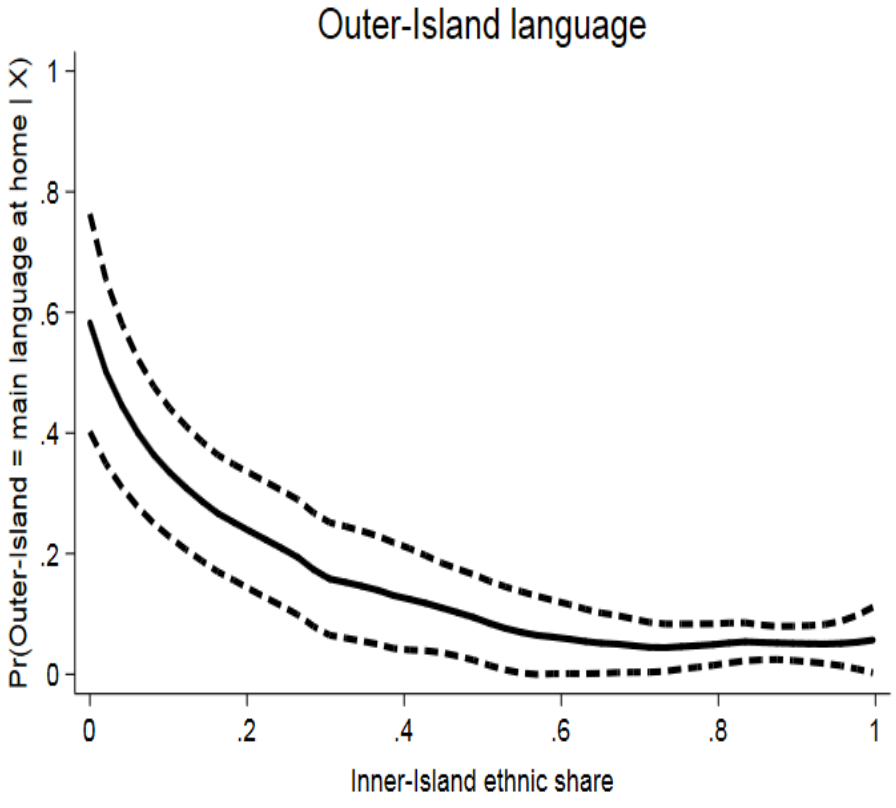
Language Use at Home by Outer-Island Ethnic Groups



LEFT: Linearity for own language, consistent w/ weakened ethnic attachment.
RIGHT: Significant inflection at $\sim$50%, consistent w/ social (network) externalities.

Ethnic Diversity and National Language Use at Home
Understanding the Nonlinear Shape $g(\cdot)$

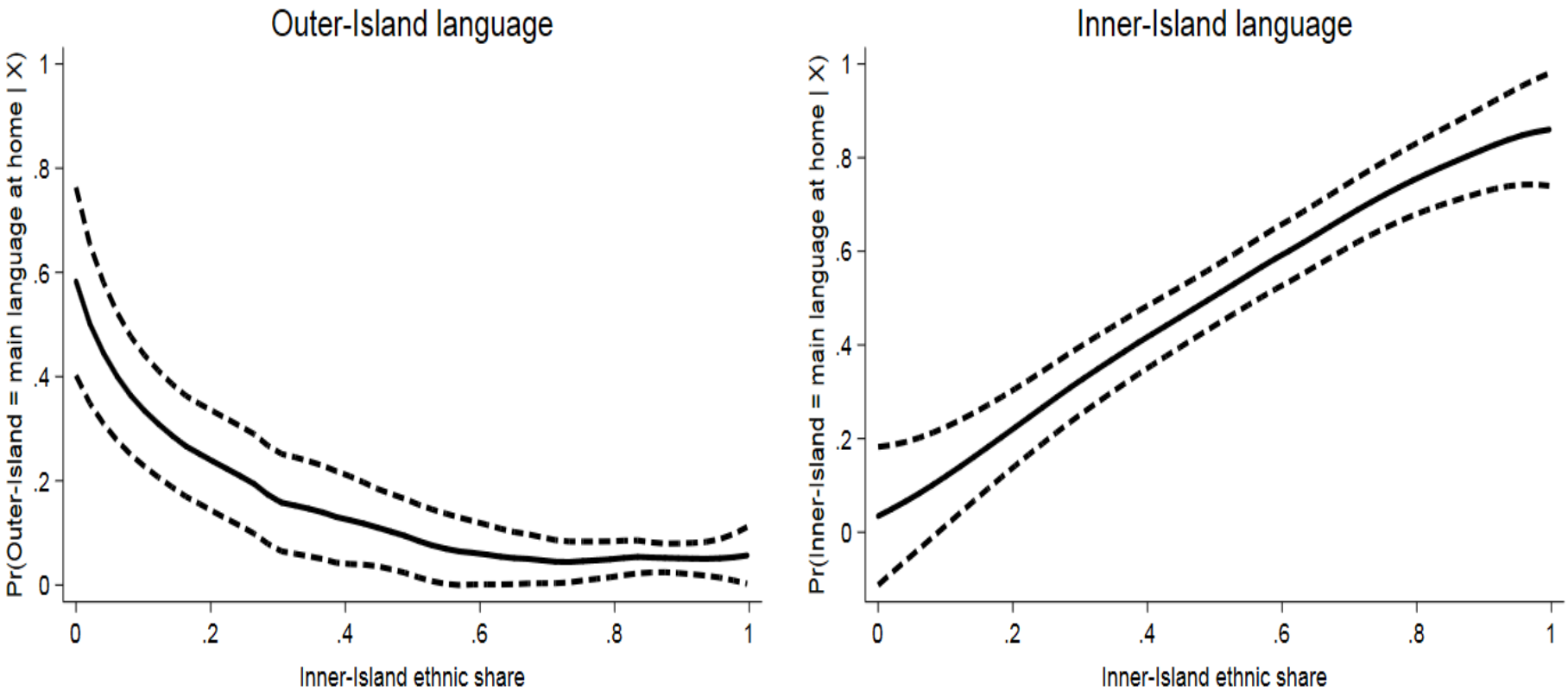
Language Use at Home by Inner-Island Ethnic Groups



LEFT: Significant inflection at $\sim$50%, consistent w/ social (network) externalities.

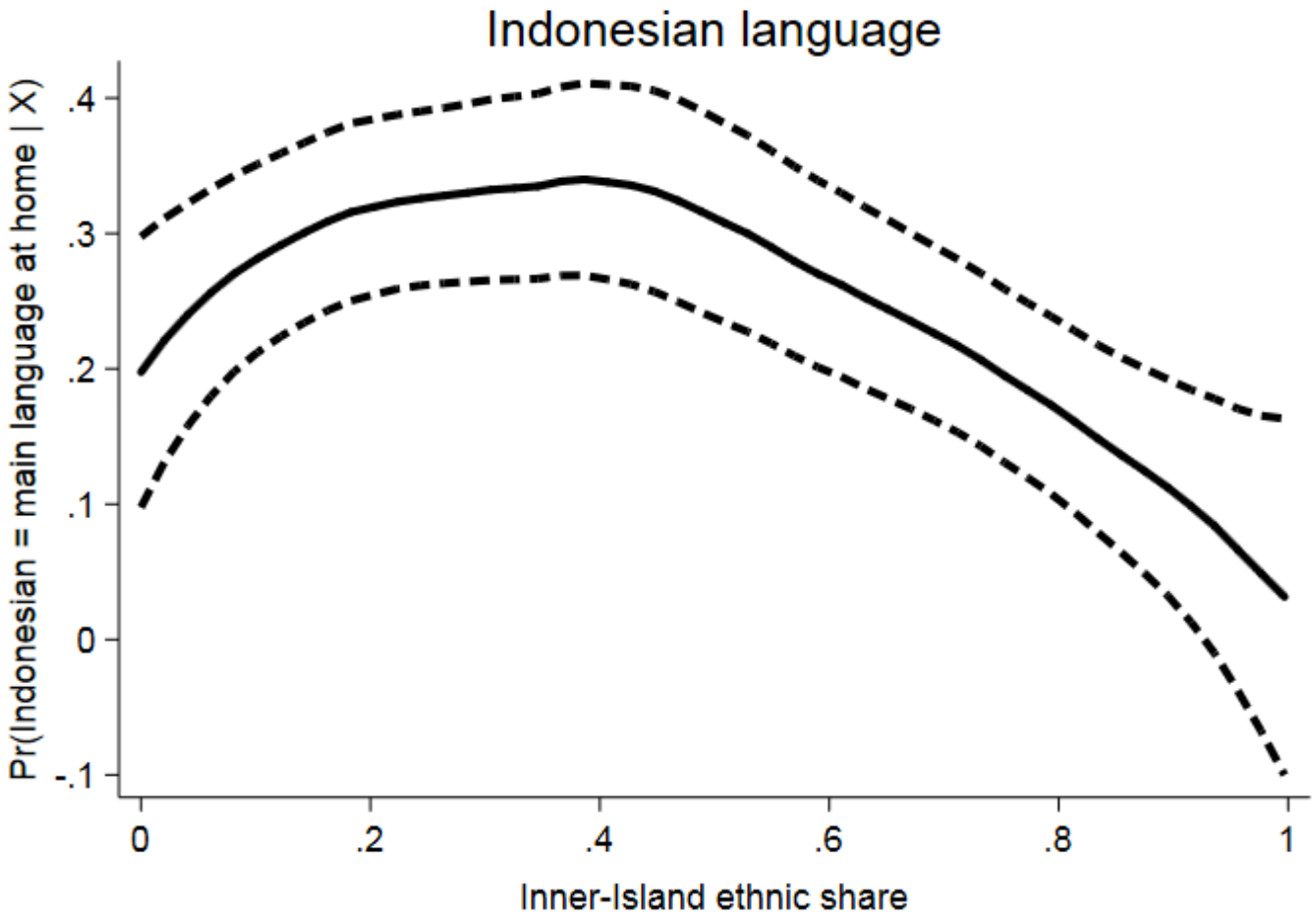
Ethnic Diversity and National Language Use at Home
Understanding the Nonlinear Shape $g(\cdot)$

Language Use at Home by Inner-Island Ethnic Groups



LEFT: Significant inflection at $\sim$50%, consistent w/ social (network) externalities.
RIGHT: Linearity for own language, consistent w/ weakened ethnic attachment.

Ethnic Diversity and National Language Use at Home



- turning point ≈ 0.4, inverted U significant at 1% (Lund & Mehlum, 2011)
- cannot reject quadratic parametric shape (Härdle & Mammen, 1993)

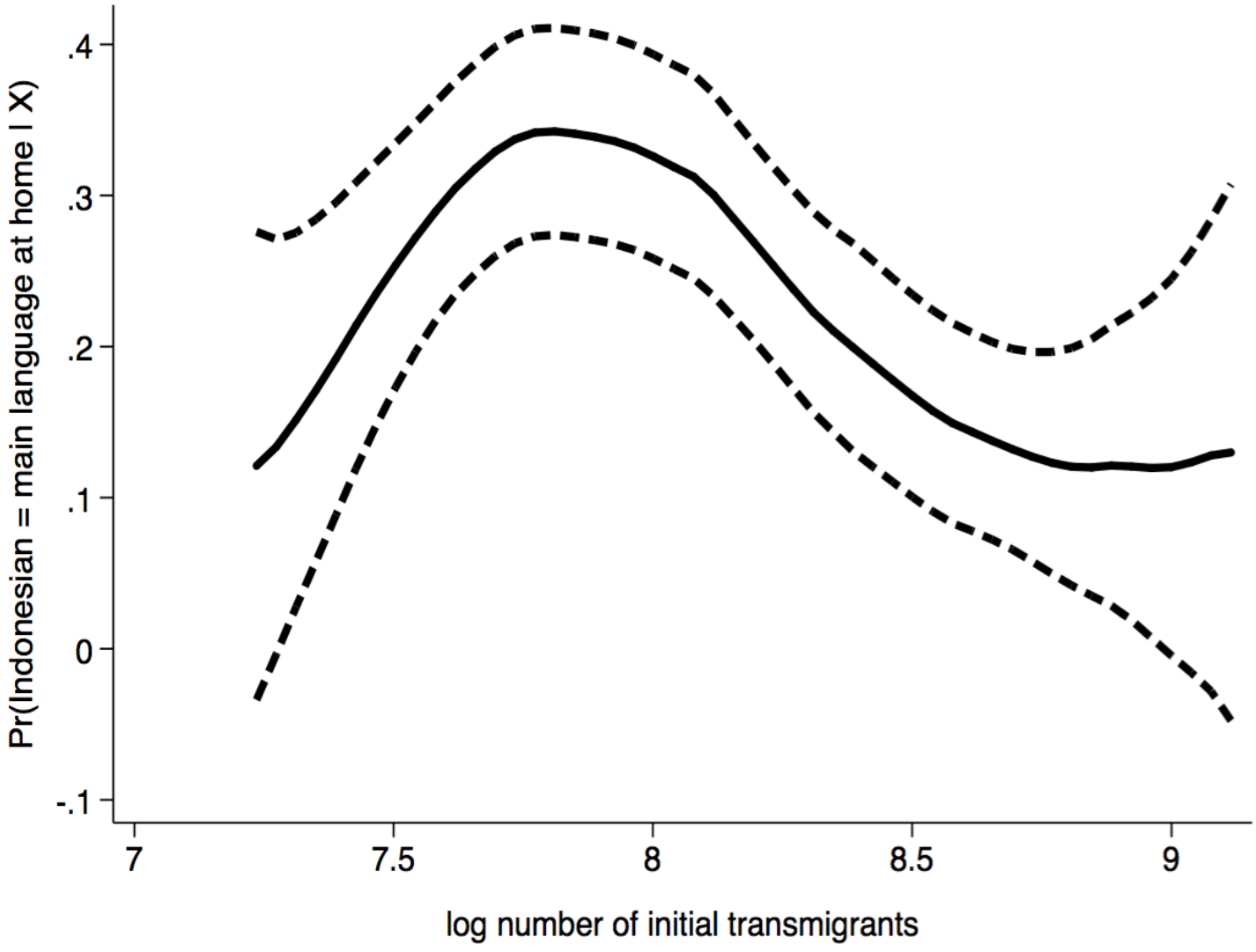
Ethnic Diversity and National Language Use at Home

Parametric OLS and IV Estimates

Estimator	OLS	IV-GMM	OLS	IV-GMM
	(1)	**(2)**	**(3)**	**(4)**
Inner-Island ethnic share	0.665	0.845		
	(0.284)**	(0.379)**		
Inner-Island ethnic share squared	-0.854	-0.927		
	(0.312)***	(0.382)**		
Inverted U Turning Point	0.390	0.456		
[p-value]	[0.012]**	[0.015]**		
Inner-Island ethnic share, bottom tercile			0.066	-0.042
			(0.058)	(0.067)
Inner-Island ethnic share, middle tercile			0.203	0.187
			(0.059)***	(0.110)*
Number of Individuals	2,126	2,126	2,126	2,126
SW Weak IV Test, Linear p-value	–	[< 0.01]	–	[< 0.01]
SW Weak IV Test, Quadratic p-value	–	[< 0.01]	–	[< 0.01]
KP Wald Stat	–	3.7	–	5.7
AR Weak Instrument Robust p-value	–	[< 0.01]	–	[< 0.01]
Hansen J Test p-value	–	[0.16]	–	[0.22]

Notes: Instruments: dummies for 20 bins of the number of initial transmigrants $<$ 10 km. Standard errors clustered by district. */**/*** denotes significant at the 10/5/1 percent significance levels. Similar results using semiparameric IV  .

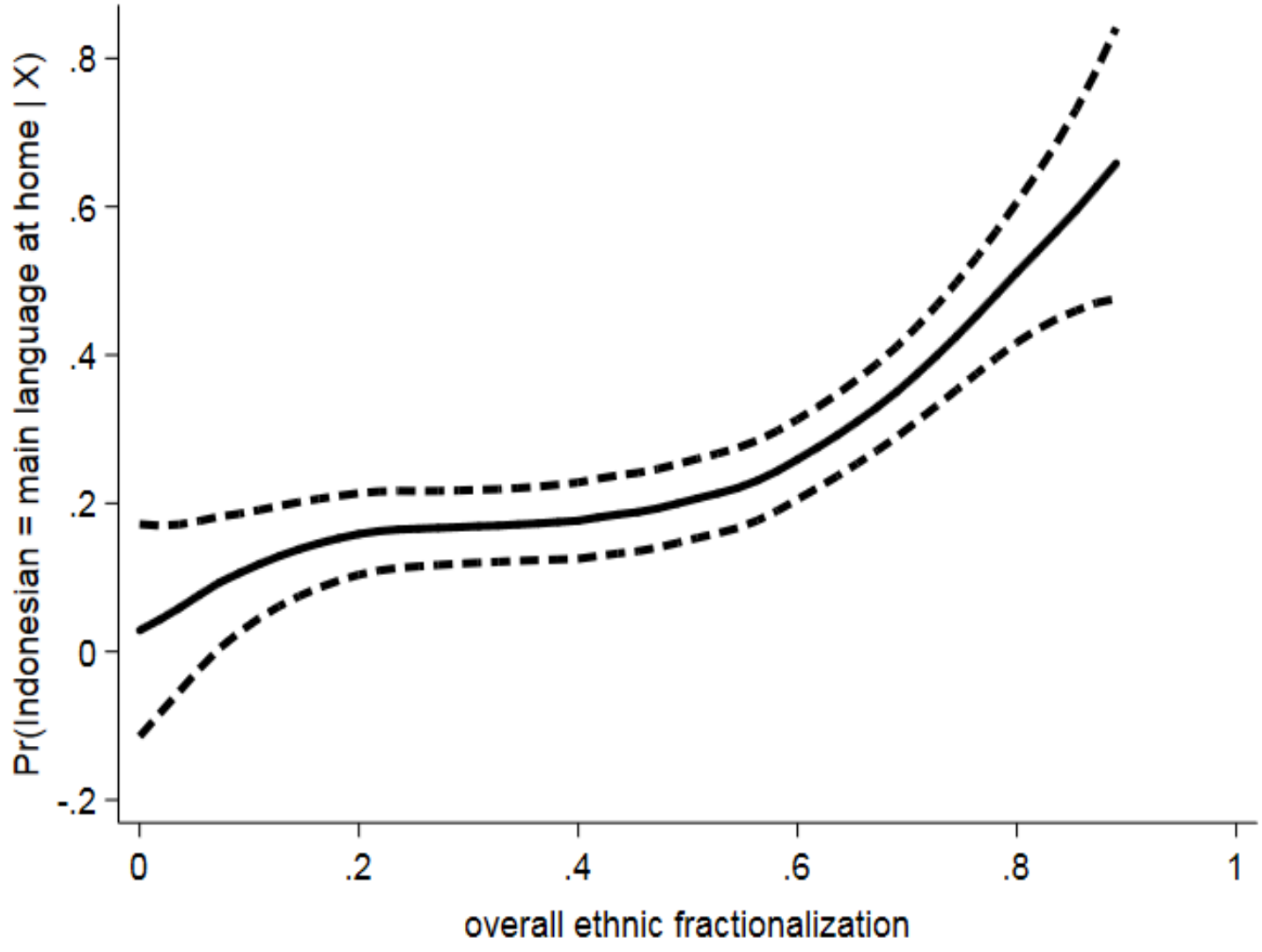
Reduced Form: Initial Transmigrants and National Language Use at Home



Robustness and Validity Checks

1. Similar results in the 1995 survey, which allows us to exclude:
   - ▷ intermarried households or spousal mother tongue mismatch
   - ▷ those w/ Indonesian mother tongue ($\Longrightarrow$ new speakers at home)
   - ▷ post-program immigrants
   - ▷ Outer-Island natives that migrated from afar in initial settlement years
2. Dropping individual controls for age and education
3. Similar shape for individuals with high and low education
4. Controlling for ethnolinguistic homeland fixed effects
5. Dropping villages with high post-program immigration
6. Oster (2017) tests for selection on unobservables ($\delta > 2$)
7. Alternative diversity: overall/Inner-Island ELF, Javanese share (latter two measures plausibly exogenous per Bazzi et al, 2016)

Overall Ethnic Fractionalization (*ELF*) and National Language Use at Home



Understanding the Potential Mechanisms

1. Scope and frequency of physical **contact with neighbors**
2. Incentives for **national integration versus majority assimilation**
3. Economic conditions: **immigrant–native skill substitutability**
4. **Ethnic political balance** and minority backlash

Understanding the Potential Mechanisms
Physical Contact with Neighbors

IV Estimates: Diversity and National Language Use at Home

	baseline + **segregation** (1)	(2)	(3)
Inner-Island ethnic share	1.421		
	(0.552)**		
Inner-Island ethnic share squared	-1.559		
	(0.583)**		
Inverted U Turning Point	0.456		
[**p-value**]	[< 0.01]***		
ethnic residential segregation (normalized index)	-0.056		
	(0.030)*		
Number of Individuals	**2,126**		

Notes: Instruments based on program-induced diversity: dummies for 20 bins of the number of initial transmigrants < 10 km. Weak-instrument diagnostics and Hansen overidentification test pass conventional significance levels across all columns. Standard errors clustered by district in parentheses. */**/*** denotes significant at the 10/5/1 percent significance.

Understanding the Potential Mechanisms
Physical Contact with Neighbors

IV Estimates: Diversity and National Language Use at Home

	baseline + segregation	distance to major historical roads	
		low	high
	(1)	(2)	(3)
Inner-Island ethnic share	1.421	0.238	1.366
	(0.552)**	(0.794)	(0.197)***
Inner-Island ethnic share squared	-1.559	-0.605	-1.584
	(0.583)**	(0.797)	(0.186)***
Inverted U Turning Point	0.456	0.197	0.431
[**p-value**]	[$<$ 0.01]***	[0.383]	[$<$ 0.01]***
ethnic residential segregation	-0.056		
(normalized index)	(0.030)*		
Number of Individuals	2,126	1,070	1,056

Notes: Instruments based on program-induced diversity: dummies for 20 bins of the number of initial transmigrants $<$ 10 km. Weak-instrument diagnostics and Hansen overidentification test pass conventional significance levels across all columns. Standard errors clustered by district in parentheses. */**/*** denotes significant at the 10/5/1 percent significance.

Understanding the Potential Mechanisms
Cultural Distance and Coordination

IV Estimates: Diversity and National Language Use at Home

	linguistic similarity w/ local homeland			
	low	high		
	(1)	(2)	(3)	(4)
Inner-Island ethnic share	0.774	0.067		
	(0.337)**	(0.142)		
Inner-Island ethnic share squared	-0.796	0.032		
	(0.381)**	(0.149)		
Inverted U Turning Point	0.486	–		
[**p-value**]	[0.034]***	–		
Number of Individuals	1,137	814		

Notes: Instruments based on program-induced diversity: dummies for 20 bins of the number of initial transmigrants < 10 km. Weak-instrument diagnostics and Hansen overidentification test pass conventional significance levels across all columns. Standard errors clustered by district in parentheses. */**/*** denotes significant at the 10/5/1 percent significance.

Understanding the Potential Mechanisms
Cultural Distance and Coordination

IV Estimates: Diversity and National Language Use at Home

	linguistic similarity w/ local homeland		**majority group ethnic diversity**	
	low	high	low	high
	(1)	(2)	(3)	(4)
Inner-Island ethnic share	0.774	0.067	0.350	1.690
	(0.337)**	(0.142)	(0.469)	(0.544)***
Inner-Island ethnic share squared	-0.796	0.032	-0.571	-1.367
	(0.381)**	(0.149)	(0.555)	(0.459)***
Inverted U Turning Point	0.486	–	0.307	0.618
[**p-value**]	[0.034]***	–	[0.230]	[0.010]**
Number of Individuals	1,137	814	1,023	992

Notes: Instruments based on program-induced diversity: dummies for 20 bins of the number of initial transmigrants < 10 km. Weak-instrument diagnostics and Hansen overidentification test pass conventional significance levels across all columns. Standard errors clustered by district in parentheses. */**/*** denotes significant at the 10/5/1 percent significance.

Understanding the Potential Mechanisms
Economic Conditions

- Agroclimatic similarity as key driver of transmigrants' productivity (Bazzi et al, 2016)
- Captures similarity in growing conditions b/t origin and destination
- As proxy for location-specific farming skills, this suggests that
  - low agroclimatic similarity $\Longrightarrow$ incentive to interact w/ native farmers
  - high agroclimatic similarity $\Longrightarrow$ more competitive with native farmers

Understanding the Potential Mechanisms
Economic Conditions

IV Estimates: Diversity and National Language Use at Home

	agroclimatic similarity	
	low	high
	(1)	(2)
Inner-Island ethnic share	1.543	-0.755
	(0.687)**	(0.146)***
Inner-Island ethnic share squared	-1.373	0.597
	(0.736)*	(0.171)***
Inverted U Turning Point	0.562	0.633 (U)
[**p-value**]	[0.073]*	[0.037]**
Number of Individuals	1,023	992

Notes: Instruments based on program-induced diversity: dummies for 20 bins of the number of initial transmigrants $<$ 10 km. Weak-instrument diagnostics and Hansen overidentification test pass conventional significance levels across all columns. Standard errors clustered by district in parentheses. */**/*** denotes significant at the 10/5/1 percent significance.

Understanding the Potential Mechanisms

Ethnic Political Balance

Understanding the Potential Mechanisms
Ethnic Political Balance

- Ethnicity is a key mobilizing force in district-level politics
- Coordination on national identity may be easier in settings where the native ethnic group is not politically threatened by transmigrant influx
- Hence, one expects less oppositional identity and greater openness to transmigrants in villages w/ majority group being politically dominant
- For example, several Transmigration settlements in ethnic Tolaki homelands of SE Sulawesi where Tolaki are majority in most districts
  - e.g., Sanuanggamo village has 54% Inner-Island ethnicity and 39% Tolaki, which comprise 65% of the district population
- In other regions, local Outer-Island native group is less dominant
  - e.g., Giri Mulya village in Bengkulu has 68% Inner-Island ethnicity and 24% Serawai, which comprise 5% of the district population

Understanding the Potential Mechanisms

Ethnic Political Balance

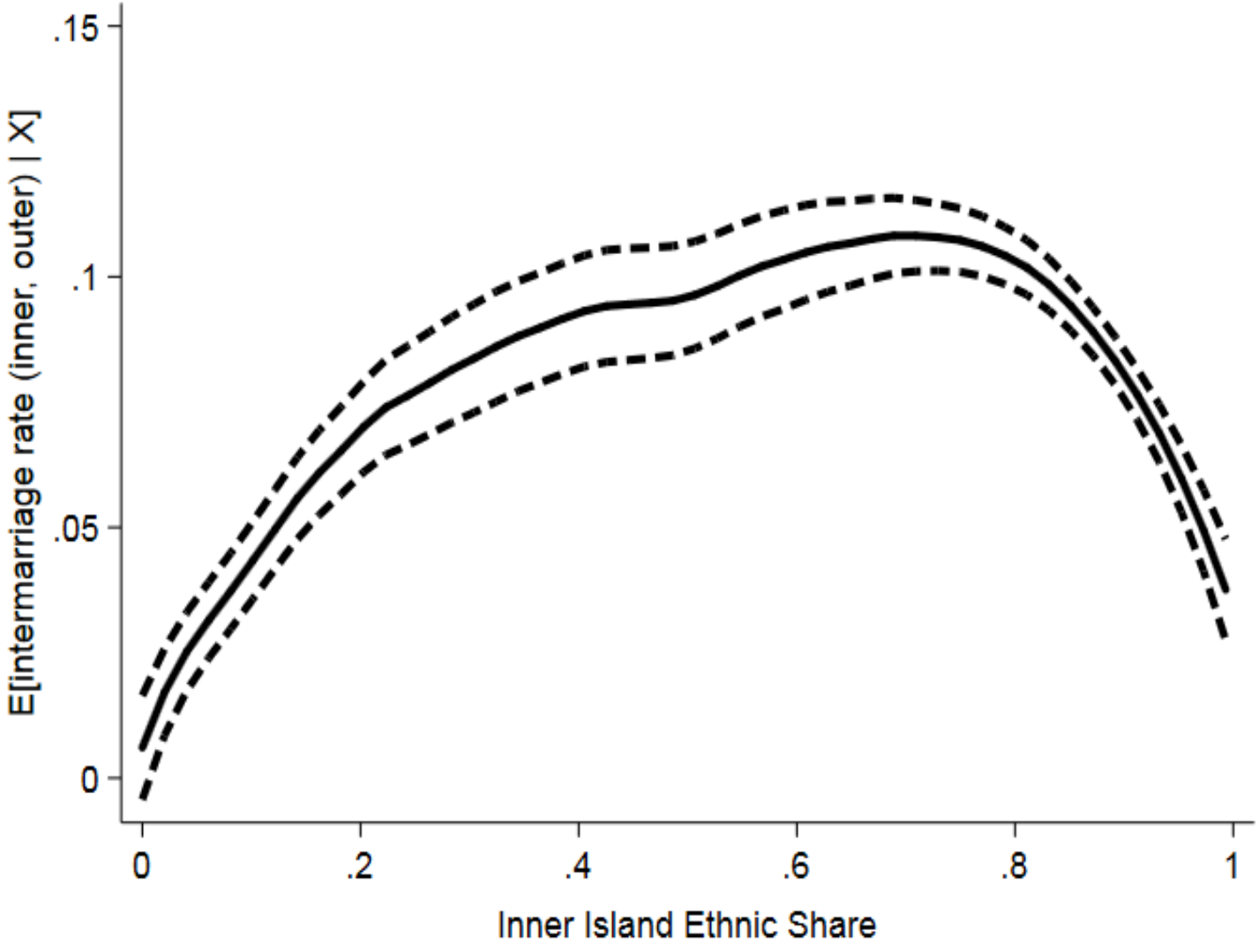
Sample Restriction	**Median Sample Splitting by Size of Largest Outer-Island Group...**					
	Within Subdistrict		Within District		Within Province	
	low	high	low	high	low	high
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Inner-Island ethnic share	-0.446	1.172	-0.593	1.306	0.869	0.850
	(0.376)	(0.327)***	(0.273)**	(0.228)***	(0.440)*	(0.198)***
Inner-Island ethnic share squared	-0.103	-1.341	-0.087	-1.233	-1.213	-0.846
	(0.382)	(0.329)***	(0.291)	(0.324)***	(0.479)**	(0.268)***
Turning point		0.437		0.530	0.358	0.502
p-value		[0.000]***		[0.007]***	[0.029]**	[0.014]**
Number of Individuals	1,072	1,054	1,071	1,055	1,071	1,055

Notes: Instruments based on program-induced diversity: dummies for 20 bins of the number of initial transmigrants < 10 km. Weak-instrument diagnostics and Hansen overidentification test pass conventional significance levels across all columns. Standard errors clustered by district in parentheses. */**/*** denotes significant at the 10/5/1 percent significance.

Diversity and Identity Formation Among the New Generation

1. **Diversity and Language Use at Home**	diversity thresholds mechanisms
2. **Diversity and Intergenerational Transmission**	intermarriage mother tongue
3. **Place-Based Impact: ATT**	$\uparrow$ diversity $\uparrow$ national language use at home $\uparrow$ intermarriage + mechanisms

Ethnic Diversity and Intermarriage
Integration Effects go Beyond Language

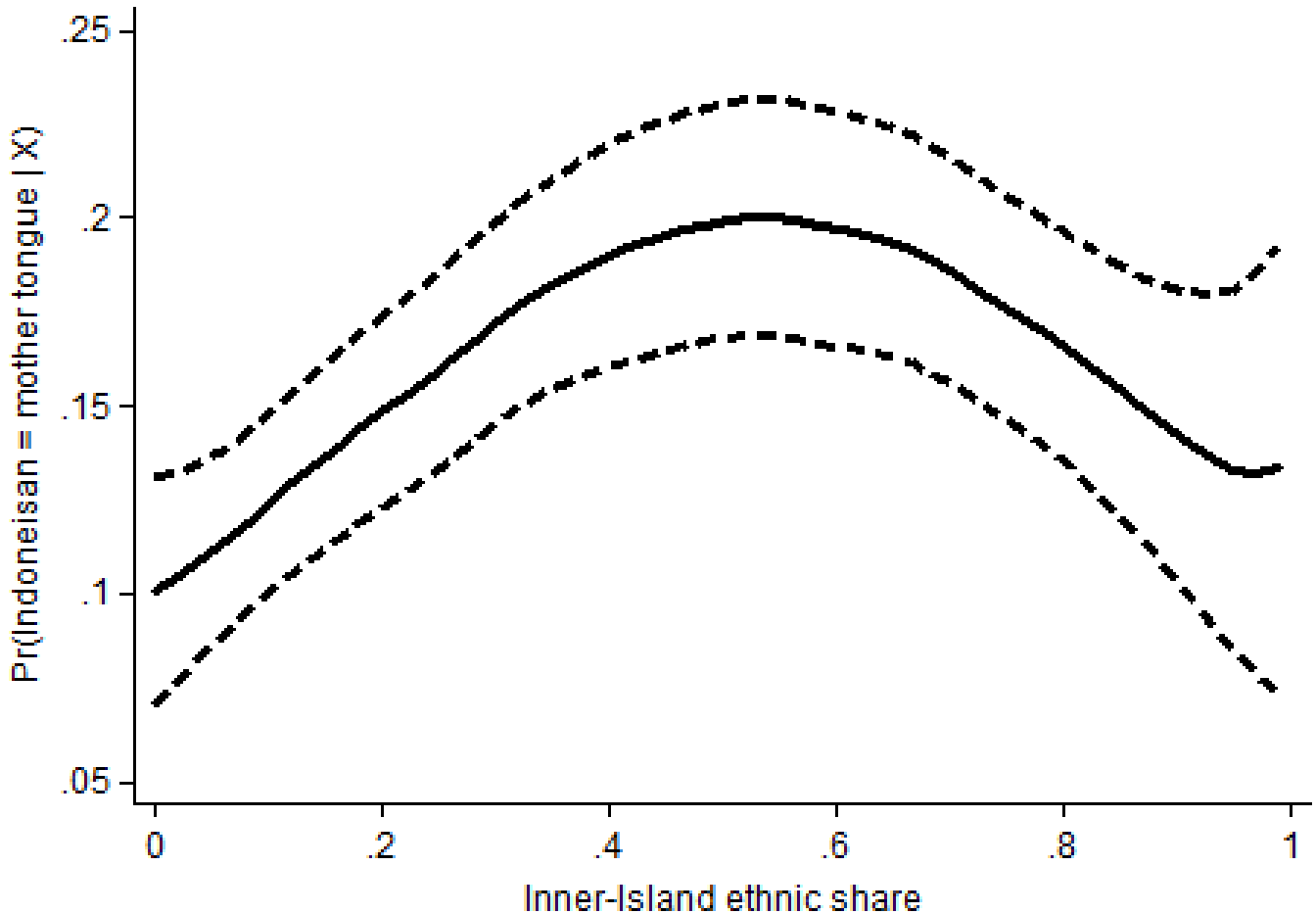


Diversity and the Transmission of National Identity

Dependent Variable: Indonesian Mother Tongue among Children

Note: mother tongue not necessarily the main language at home

$\Longrightarrow$ mother tongue captures fluid measure of cultural identity



Notes: Robinson (1988) partially linear OLS regression.

Diversity and the Transmission of National Identity

Dependent Variable: Indonesian Mother Tongue among Children

	(1)	(2)	(3)	(4)
Inner-Island ethnic share	2.313	2.095	1.072	0.927
	(0.525)***	(0.528)***	(0.327)***	(0.321)***
Inner-Island ethnic share squared	-2.372	-2.127	-1.020	-0.860
	(0.554)***	(0.561)***	(0.344)***	(0.341)**
Turning point	0.488	0.492	0.525	0.539
p-value	[0.000]***	[0.000]***	[0.006]***	[0.018]**
parents intermarried		0.197		0.142
		(0.039)***		(0.025)***
father has Indonesian mother tongue			0.401	0.394
			(0.026)***	(0.025)***
mother has Indonesian mother tongue			0.456	0.449
			(0.033)***	(0.032)***
Number of Individuals	13,325	13,325	13,325	13,325
Dependent Variable Mean	0.158	0.158	0.158	0.158

Diversity and the Transmission of National Identity

Dependent Variable: Indonesian Mother Tongue among Children

	(1)	(2)	(3)	(4)
Inner-Island ethnic share	2.313	2.095	1.072	0.927
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			(0.033)***	(0.032)***
Number of Individuals	13,325	13,325	13,325	13,325
Dependent Variable Mean	0.158	0.158	0.158	0.158

Results hold when restricting to children whose mother does not report Indonesian as her main language spoken at home

$\Longrightarrow$ even stronger measure of national identity among next generation

Roadmap

Introduction

Background: Diversity and the Transmigration Program

Data: Diversity and Nation Building Outcomes

Diversity, Socialization, and Identity Formation

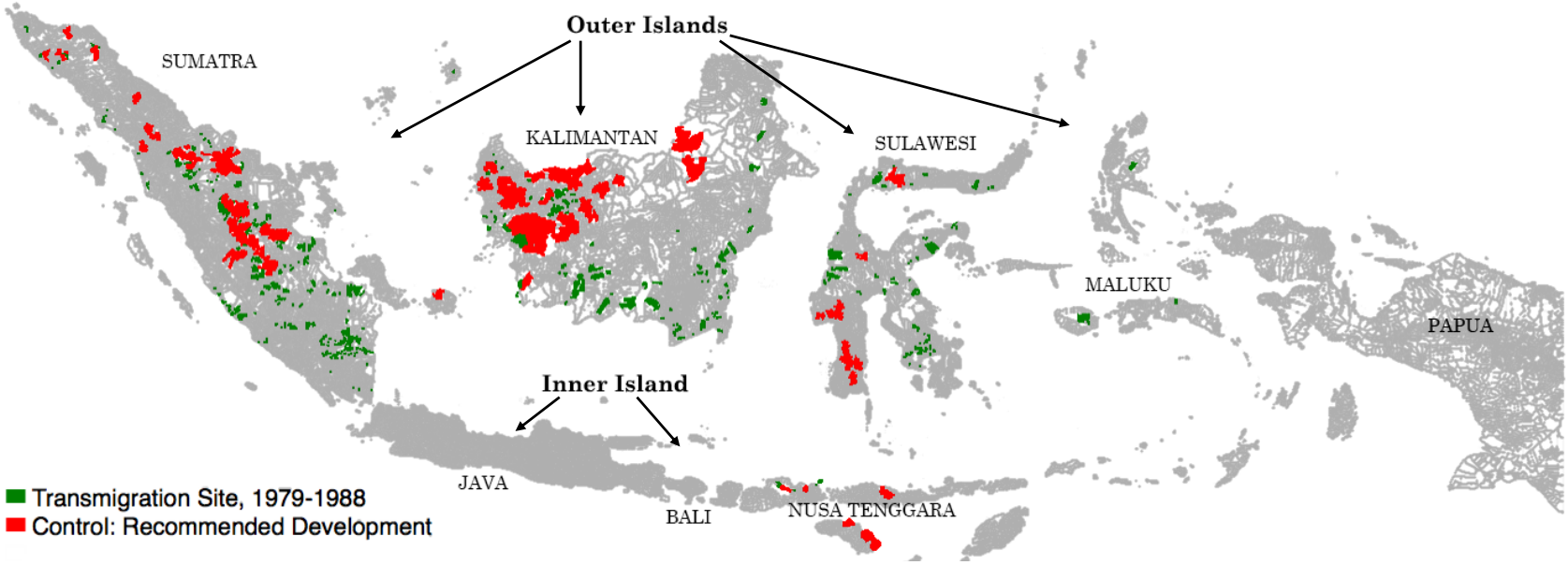
Did the Transmigration Program Foster Nation Building?

Discussion

A Natural Policy Experiment in Ethnic Mixing

Transmigration: voluntary rural-to-rural resettlement, 1979–1988

- 2 mn. migrants from Java/Bali placed in >**900 new villages**
- each community contained a **mix of Inner and Outer Islanders**
- goals of population redistribution + integration, but very controversial



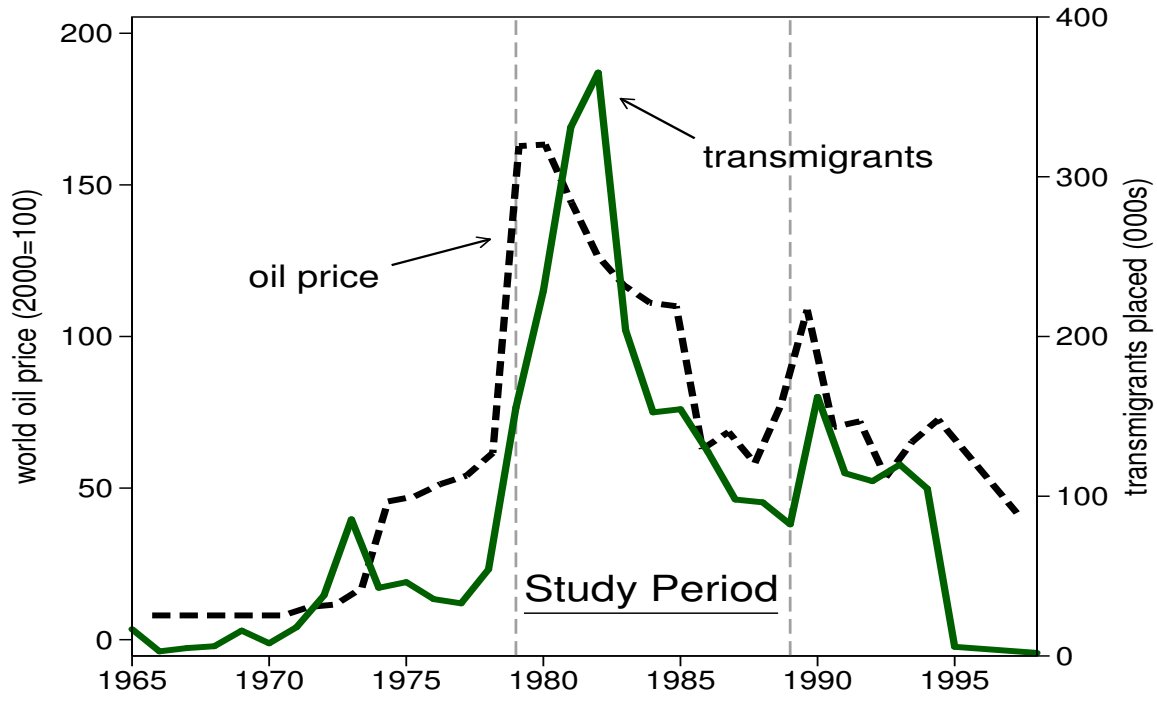
Oil boom/bust $\Longrightarrow$ **policy discontinuity** $\Longrightarrow$ **counterfactual sites**

- **treatment**: state-sponsored settlement of new villages
- **control**: spontaneous settlement of new villages around same time

Demographic Shocks and Cultural Change

1. **Diversity and Language Use at Home**	diversity thresholds mechanisms
2. **Diversity and Intergenerational Transmission**	intermarriage mother tongue
3. Place-Based Impact: ATT	↑ diversity ↑ national language use at home ↑ intermarriage + mechanisms

Rapid Scale Up and Sudden Contraction



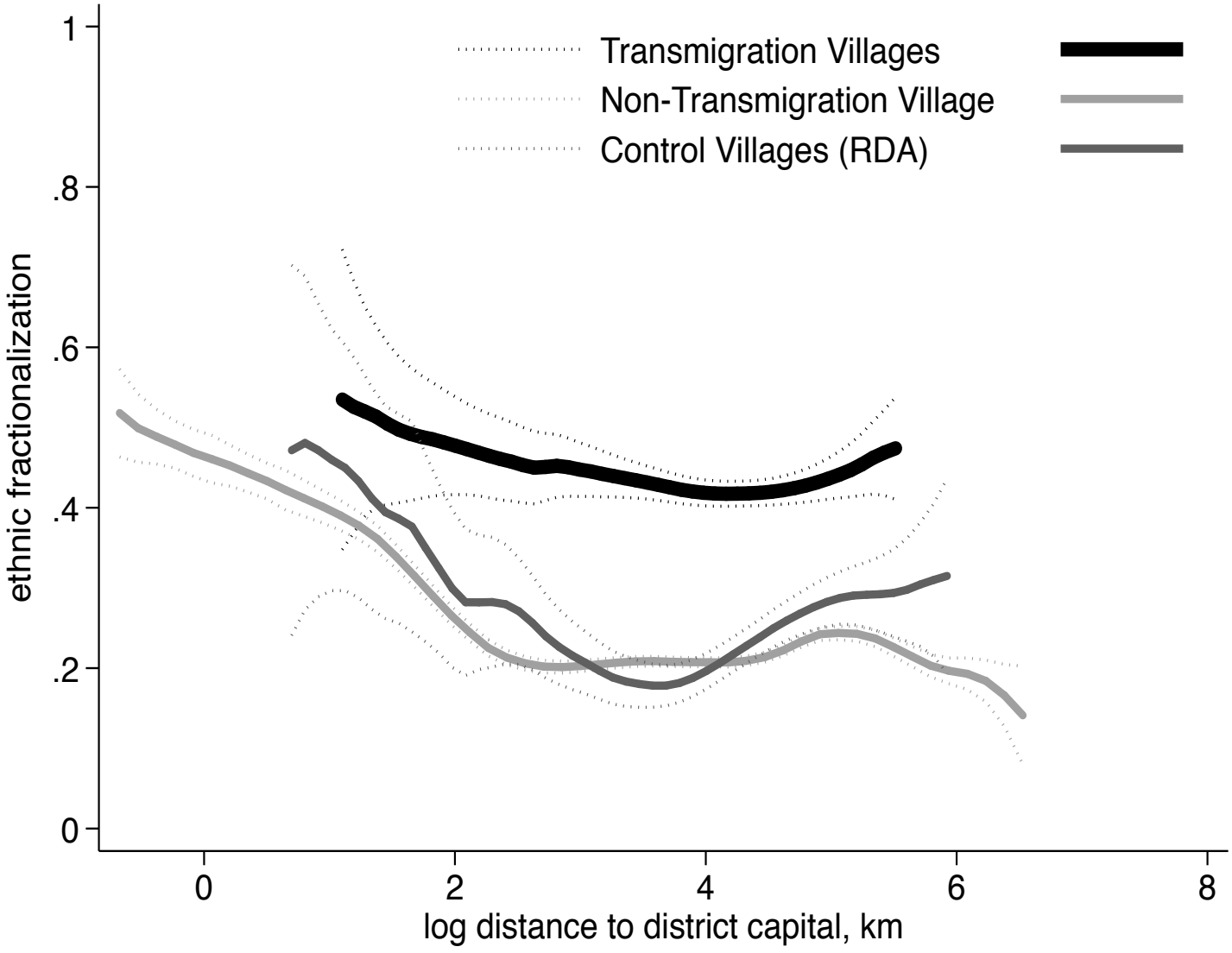
"Oil Bust"

- initial target: 2.5 mn. people in 79–83, and 3.75 mn. in 84–88
- budget cut: Rp. 578 bn (FY85/86) $\rightarrow$ Rp. 325 (FY86/87)

  $\Longrightarrow$ only 2 million transmigrants settled

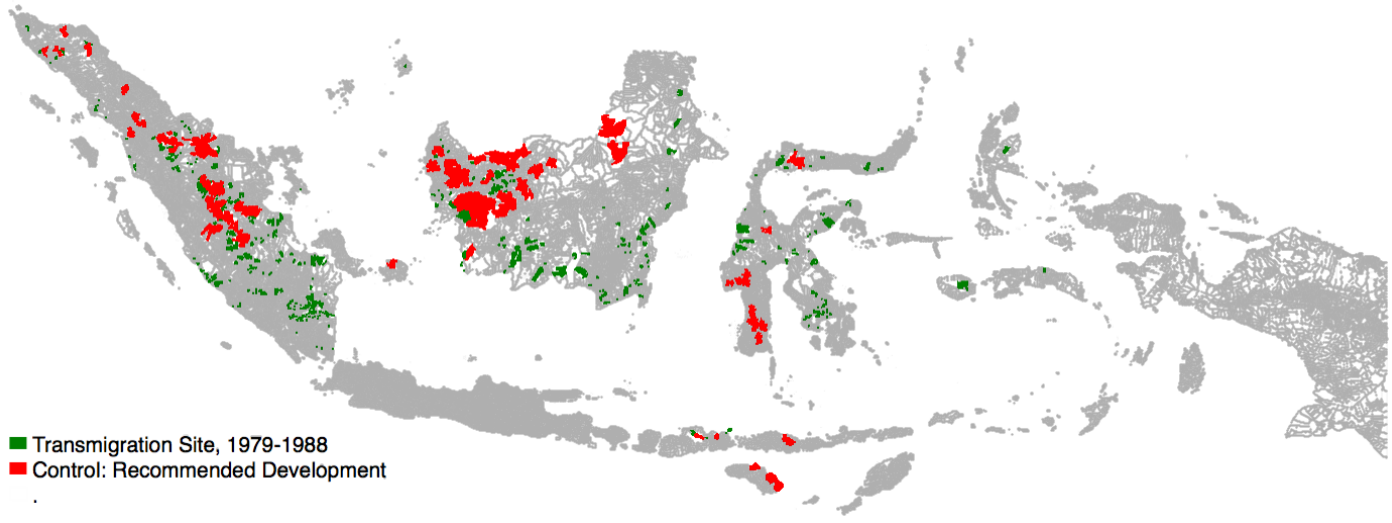
  $\Longrightarrow$ **many planned sites (RDAs) did not receive transmigrants**

Rapid Scale Up and Sudden Contraction



What Was the Impact of Transmigration on the Outer Islands?

Identifying Program Impacts



Oil price $\downarrow \implies$ policy discontinuity $\implies$ counterfactual new villages

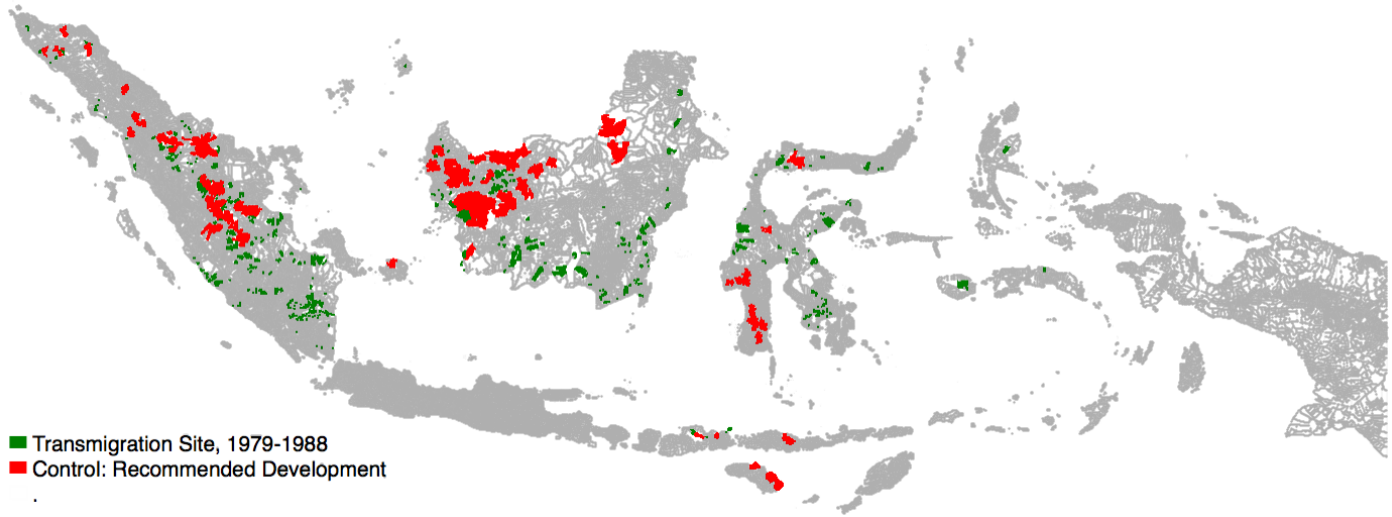
$$y_j = \alpha + \beta \text{Transmigration}_j + \mathbf{x}_j'\boldsymbol{\beta} + \nu_j$$

where $\text{Transmigration}_j = 1$ if treated, $= 0$ if control (RDA) ▸ RDA example

- 832 treated villages, 668 control villages ($>$ 10km from treated)
- $\mathbf{x}_j$: predetermined site selection variables

What Was the Impact of Transmigration on the Outer Islands?

Identifying Program Impacts



Oil price $\downarrow \implies$ policy discontinuity $\implies$ counterfactual new villages

$$y_j = \alpha + \beta \text{Transmigration}_j + \mathbf{x}_j'\beta + \nu_j$$

- Place-based evaluation: **reweight control villages** by $\widehat{\text{odds}}(T_j = 1)$
  (Blinder-Oaxaca Double Robust: Kline, 2011; Kline and Moretti, 2014; Busso et al, 2013)

$\implies$ **balanced natural advantages** between treatment & control
$\implies$ similar counterfactual demographics and initial institutions

▸ table ▸ intuition

Village-Level Demographic Impacts

population $\sim$, diversity $\uparrow$

Dependent Variable	ATT	Control Group Mean
log population	0.068 (0.088)	7.2
Inner Island-born population share	0.335 (0.019)***	0.020
Inner-Island ethnicity share	0.540 (0.038)***	0.061
ethnic fractionalization	0.126 (0.031)***	0.238

Notes: All regressions based on the Blinder-Oaxaca reweighting approach with **x** and island fixed effects. Standard errors clustered by district in parentheses. */**/*** denotes significance at the 10/5/1 percent level.

Transmigration $\Longrightarrow$ National Language Use at Home
Individual-Level ATT Estimates

	Dependent Variable:		
	$\mathbb{P}$**(Daily ... Language Use at Home)**		
	Indonesian	**Inner Island**	**Outer Island**
	(1)	(2)	(3)
1. Baseline ATT	0.250	-0.002	-0.248
	(0.126)**	(0.068)	(0.162)
Number of Individuals	2,878	2,878	2,878
Control Group Mean	0.122	0.073	0.805

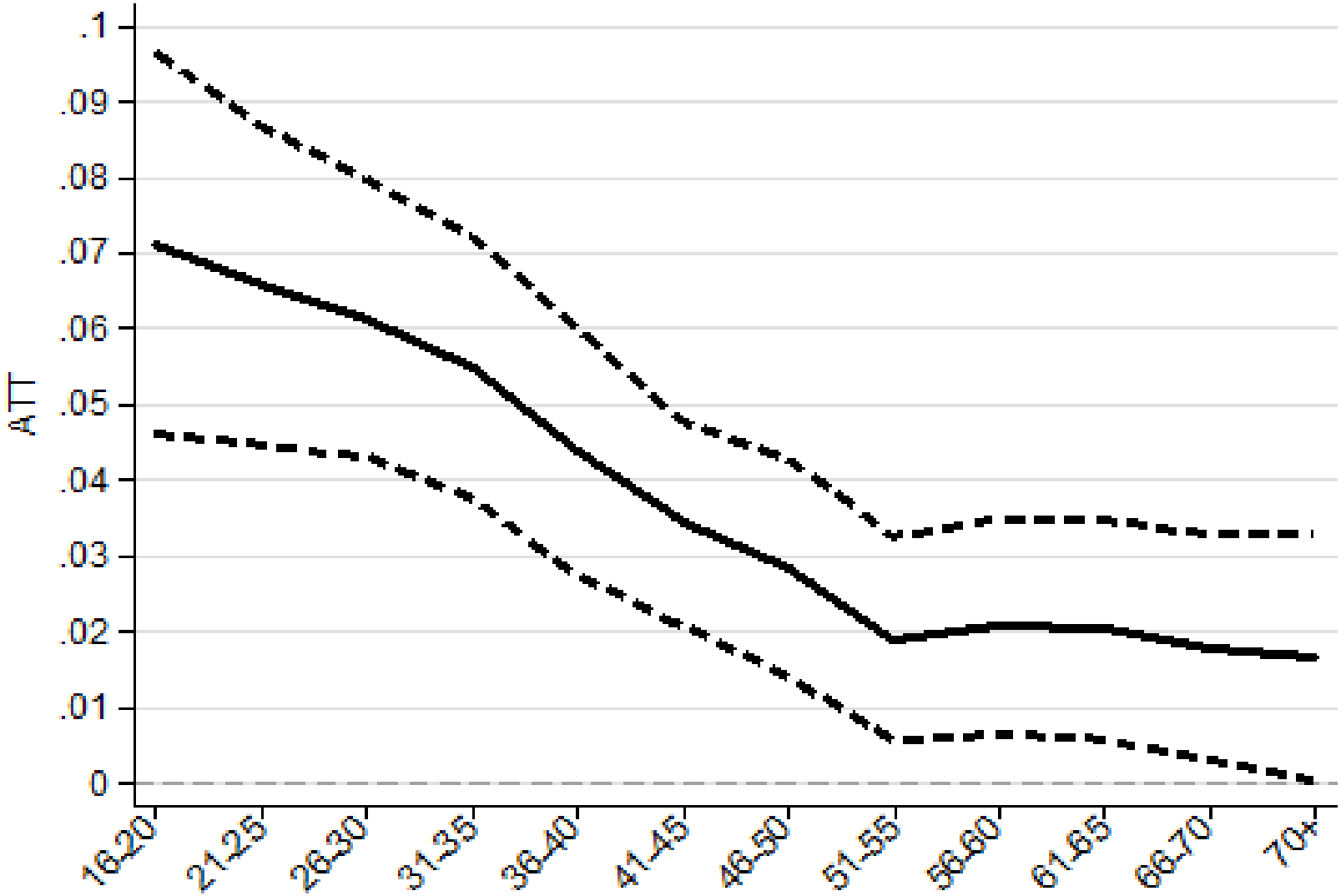
- **effect size** $\approx$ differential Indonesian use among (1) middle school (college) vs. none (middle school), (2) urban vs. rural households
- robust to battery of additional controls ▸ table
- similar effect sizes in 1995 survey data but lower mean use at home
- results driven by exposed, non-immigrant Outer Islanders

Individual-Level Age-Specific ATT for Intermarriage

Restricting to Local Native-Born Outer-Island Ethnics

Individual-Level Age-Specific ATT for Intermarriage

Restricting to Local Native-Born Outer-Island Ethnics



Notes: Include birth district and ethnicity fixed effects. 95% confidence interval on age group-specific ATT estimates.

Ruling out Confounders of Ethnicity in Treated vs. Control

- **Similar ATT** for Outer-Island **men and women**
  $\Longrightarrow$ no differential intermarrying for land acquisition (among others)

- **Similar ATT** for **high and low education** Outer Islanders
  $\Longrightarrow$ no differential intermarrying for (lack of) assortativity

- **Similar ATT** for Outer Islanders in **trade & non-trade occupation**
  $\Longrightarrow$ no differential intermarrying for economic exchange purposes

. . . **same results with FE for years of schooling and occupation**

  - also, no differential effects on marriage rates

- **Similar patterns and results for any interethnic marriage**

Aggregate Intermarriage Impacts for Younger Cohort

Dependent Variable	ATT	Control Group Mean
marriage rate	0.019 (0.013)	0.829
intermarriage rate	0.050 (0.006)***	0.023

- **effect size $\approx$ intermarriage gap between education levels** (eg, primary vs none)
- **Δ supply of non-coethnics explains $\approx$1/3rd overall effect**

▸ reduced form decomposition

- **robust to**: (1) province FE, (2) linguistic homeland FE, (3) other predetermined individual/location **x**, (4) INPRES school construction, (5) Oster (2017) test

Roadmap

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Discussion

Key Takeaways

A Policy Experiment in Ethnic Mixing

- Diversity $\Longrightarrow$ integration/socialization consistent w/ nation building
  - evidence contrary to popular claims about program and conflict (*but* consistent w/ recent reappraisal by Barter & Cote, 2015)

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  (ii) mind the broader political context and ethnic balance

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- Policy: (i) opportunities + incentives for cooperation amid Δ diversity
  (ii) mind the broader political context and ethnic balance

- Local diversity can support nation building via externalities of marriage and socialization decisions such as language use at home

- Important given (ethnic) segregation $\Longrightarrow$ adverse aggregate policy
  (homogenous communities w/ high social capital but weak national integration)

- Nevertheless, ethnic inequality may undermine benefits of contact

External Validity: Broader Relevance

Public Policy around Interethnic Cooperation

- Migration and **resettlement** pressures rising globally
  - resettlement policy challenge due to conflict, climate change, etc (de Sherbenin et al, 2011)
- What role for **state-sponsored internal migration** given many examples outside Indonesia of less benign intentions and outcomes?
  - growing evidence: spontaneous migration $\Longrightarrow$ 'sons of the soil' conflict
- **Integration policies** in OECD countries w/ growing immigration
- **Language policy**: national vs. official vs. majority
  - National language in Indonesia compared to India and Philippines
  - Success of Swahili in Tanzania
  - French as unifying language in historical France
  - Ongoing debates in Spain, Sri Lanka, ...

APPENDIX

Examples of Inner–Outer Group Differences

based on interethnic marriages observed in study villages

1. **Javanese vs. Batak (North Sumatra)**
   - ▷ Batak have bride price while Javanese typically do not
   - ▷ Batak are patrilocal while Javanese are matrilocal
   - ▷ Batak have patrilineal inheritance while Javanese have equal inheritance

2. **Javanese vs. Minang (West Sumatra)**
   - ▷ Minang and Javenese have no marital wealth exchange traditions
   - ▷ Minang have no common post-marital residence rules while Javanese are matrilocal
   - ▷ Minang have matrilineal inheritance while Javanese have equal inheritance

3. **Balinese vs. Toraja (Central Sulawesi)**
   - ▷ Toraja have bride price while Balinese typically do not
   - ▷ Toraja are matrilocal while Balinese are patrilocal
   - ▷ Toraja have equal inheritance while Balinese have patrilineal inheritance

Generally, Inner–Inner differences dwarfed by Inner–Outer differences

▸ back

Examples of Inner–Outer Group Differences

based on interethnic marriages observed in study villages

Linguistic Differences

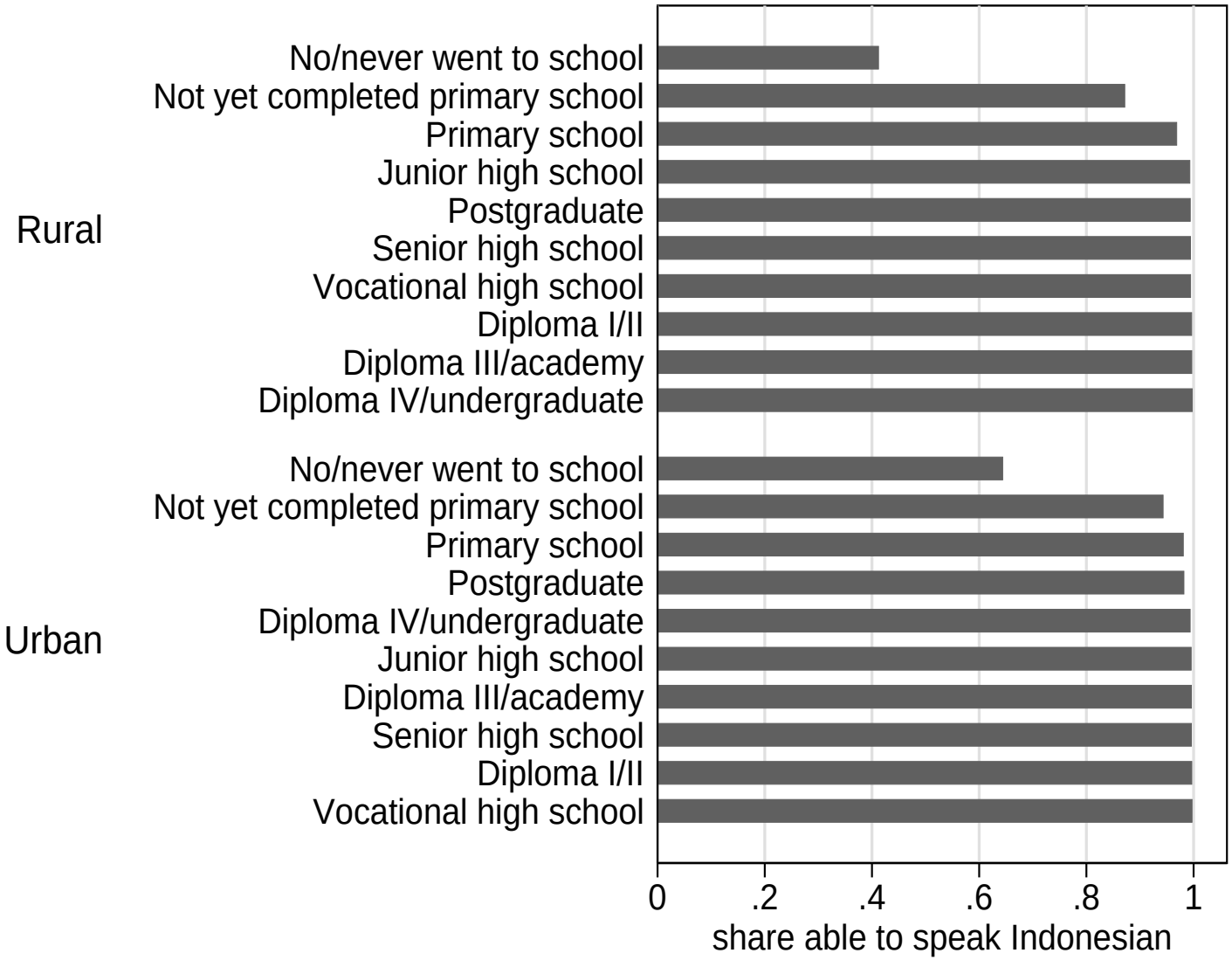
	Branches						
Language	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Javanese	Austronesian	Malayo-Polynesian	Javanese				
Minangkabau	Austronesian	Malayo-Polynesian	Malayo-Sumbawan	North and East	Malayic	Malay	
Batak	Austronesian	Malayo-Polynesian	NW Sumatra-Barrier Islands	Batak	Southern		
Toraja	Austronesian	Malayo-Polynesian	South Sulawesi	Northern	Toraja-Sa'dan		
English	Indo-European	Germanic	West	English			
German	Indo-European	Germanic	West	High German	German	Middle German	East Middle German
French	Indo-European	Italic	Romance	Italo-Western	Western	Gallo-Iberian	Gallo-Romance

Notes: Ethnologue language classification.

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National Language Ability and Use at Home (2010 Census)

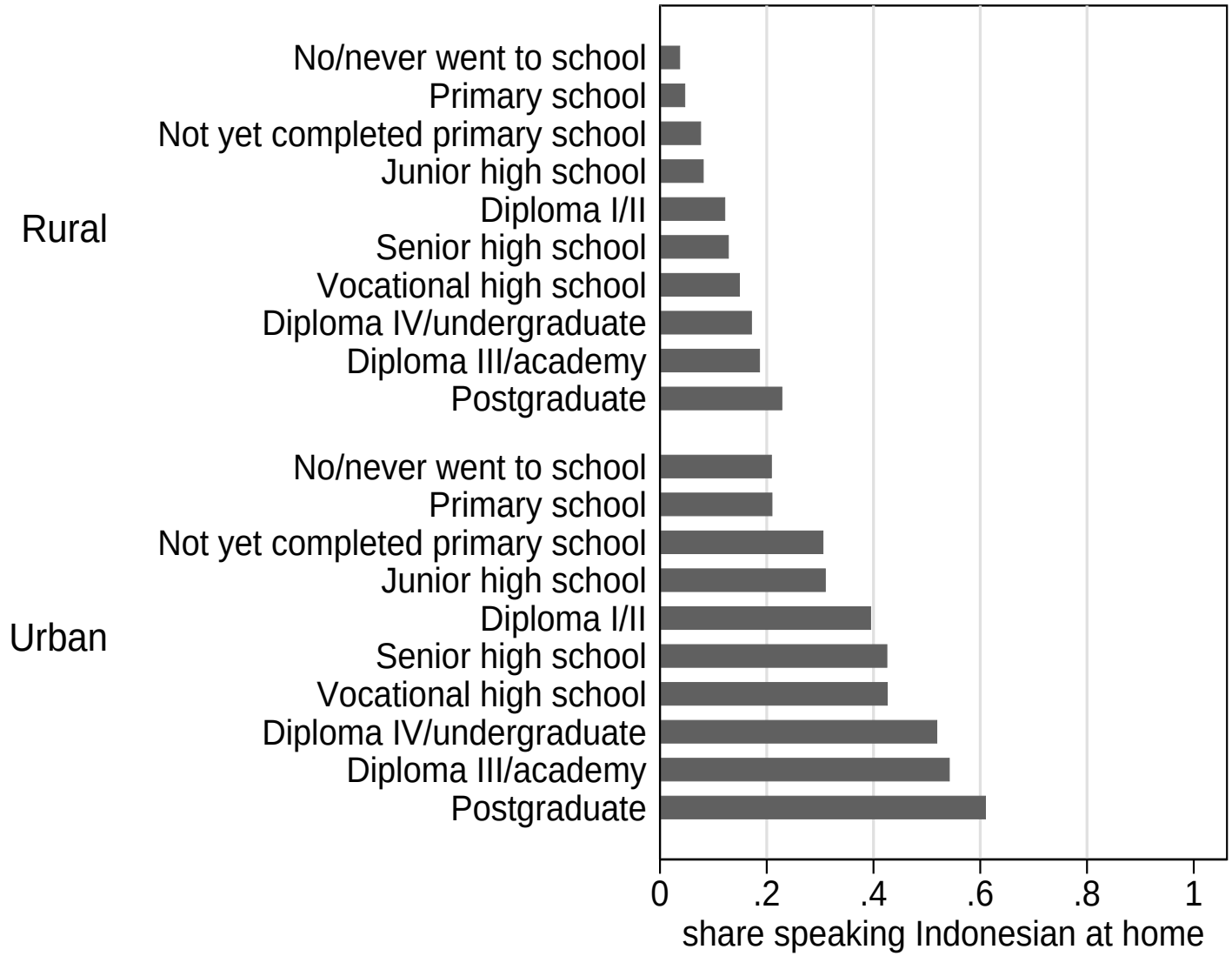
Share Able to Speak Indonesian



back

National Language Ability and Use at Home (2010 Census)

Share Speaking Indonesian as the Main Language at Home



back

Transmigrants Are Slightly Negatively Selected

	Years of Schooling Relative to Java/Bali-born Stayers in Transmigration-eligible Cohort			
	2000 Census		1985 Inter-Census	
	(1)	**(2)**	**(3)**	**(4)**
Migrant to Transmigration site	-0.650	-0.731	-1.179	-1.044
	(0.136)***	(0.088)***	(0.272)***	(0.229)***
Migrant to other Outer Islands rural area	3.267	2.407	3.272	2.600
	(0.122)***	(0.087)***	(0.256)***	(0.368)***
Migrant to other Outer Islands urban area	4.057	3.186	3.672	3.134
	(0.127)***	(0.111)***	(0.168)***	(0.216)***
Migrant to Java/Bali rural area	-0.212	-0.411	-1.014	-0.924
	(0.140)	(0.093)**	(0.187)***	(0.141)***
Migrant to Java/Bali urban area	3.762	2.652	2.709	2.138
	(0.177)***	(0.149)***	(0.278)***	(0.276)***
Number of Individuals	41,201,749	41,201,749	39,766,326	39,766,326
Age FE	No	Yes	No	Yes
Birth District FE	No	Yes	No	Yes

Regression of years of schooling on mutually exclusive dummy variables indicating type of migrant with non-migrants as the reference. Standard errors clustered at the district level. */**/*** denotes significance at the 10/5/1 percent level.

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Summary Stats: Transmigration Villages

back way back

	Mean	Std. Deviation
(a) Demographics and Residence		
total population	2,010	(1,306)
Inner-Island born population share	0.38	(0.20)
Inner-Island ethnic share	0.67	(0.31)
ethnic fractionalization	0.42	(0.21)
number of ethnic groups	23.65	(20.18)
ethnic residential isolation index (inner, outer)	0.15	(0.18)
(b) Primary Language at Home		
All Individuals		
National Language	0.25	(0.43)
Outer-Island Language	0.46	(0.50)
Inner-Island Language	0.29	(0.46)
Inner-Island Natives		
National Language	0.26	(0.44)
Outer-Island Language	0.08	(0.26)
Inner-Island Language	0.66	(0.47)
Outer-Island Natives		
National Language	0.24	(0.43)
Outer-Island Language	0.74	(0.44)
Inner-Island Language	0.02	(0.14)
(c) Marriage among Young Cohort		
marriage rate	0.85	(0.14)
intermarriage rate (inner, outer)	0.08	(0.07)
adjusted intermarriage rate (inner, outer)	0.39	(0.37)

Notes: This table reports summary statistics for the 832 Transmigration villages and 668 control villages in our baseline

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II Probing Instrument Validity

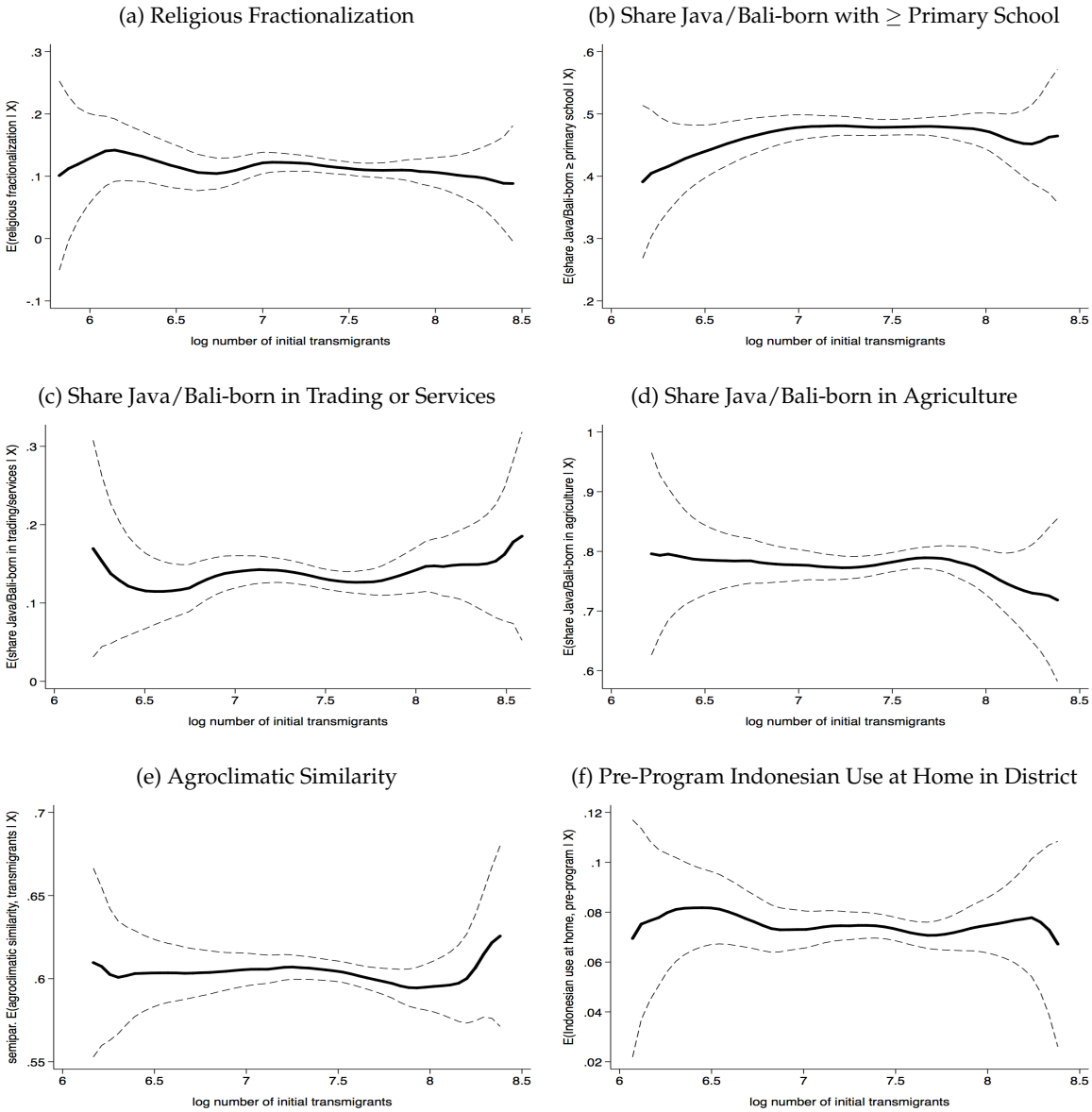
This section describes results mentioned in Section 5 supporting the excludability of the initial number of transmigrants as an instrument for ethnic diversity in 2000. We show that there is not a systematic inverted-U relationship between the initial transmigrant assignment and a wide array of omitted variables potentially correlated with ethnic diversity and integration. These results help rule out confoundedness insomuch as the lack of an inverted U stands in contrast to the strong reduced form, inverted-U relationship in Figure 6.

First, as seen in panels (a)-(d) of Appendix Figure A.3, the instrument does not predict other measures of population diversity such as religious fractionalization, skill levels, or occupational mixes among the initial transmigrants. This suggests that larger settler groups were not mechanically more likely to have greater diversity along other dimensions besides ethnicity. Second, the instrument is uncorrelated with other predetermined proxies for political and economic development not captured in the $\mathbf{x}$ vector used for site selection (see Appendix Figure A.4). These proxies include measures of potential agricultural yields, malaria suitability in 1978, the district-level share of votes going to the Golkar party of President Suharto in the 1977 legislative elections, and a host of district-level characteristics of the population residing within these areas (but not in the immediate settlements) as of 1978, including information on wealth, infrastructure access, schooling, and sector of work.

Additionally, panel (e) Appendix Figure A.3 shows a flat relationship with agroclimatic similarity (see Bazzi et al., 2016), suggesting that the assignment rule was not systematically correlated with the transferability of skills among the pool of potential transmigrants at a given point in time. Finally, panel (f) Appendix Figure A.3 shows a similar flat relationship with respect to national language use at home in the late 1970s in areas near the eventual Transmigration settlements but prior to their creation. The lack of an inverted-U relationship here rules out the concern that planners created more diverse settlements in districts with already high levels of national affinity.

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Figure A.3: Probing the Validity of the Initial Transmigrant Assignment



Notes: This figure reports semiparametric Robinson (1988) estimates of the relationship between the number of initial transmigrants instrument and other potential confounders of our main diversity measure, the Inner-Island ethnic share. These graphs serve to rule out first order concerns that the instrument for the Inner Island ethnic share (see Figure 3) is correlated with other measures of diversity and population characteristics associated with the initial immigrant influx. We capture in panel (a) religious diversity in the village, (b) the share of transmigrants (born in Java/Bali) with at least primary school, (c) the share of transmigrants working in trading or services, (d) the share of transmigrants in agriculture, (e) the agroclimatic similarity between transmigrants' origin and the given destination, which is a strong proxy for their economic well-being (see Bazzi et al., 2016), and (f) the share of the district that spoke the national language at home in 1978 based on the population residing in the given village's district at the time according to the 1980 Population Census. The semiparametric regression is based on a local linear regression that conditions on island fixed effects and the vector $\mathbf{x}$ of predetermined site selection variables, an Epanechnikov kernel and Fan and Gijbels (1996) rule-of-thumb bandwidth. The top and bottom p percentiles of the x-axis are trimmed for presentational purposes where p varies across figures but is in the 0–4 range.

Robinson (1988) Double Residual Estimator

We want to estimate the shape of $m(\cdot)$:

$$y_j = \alpha + m(\mathit{diversity}_j) + \mathbf{x}_j'\boldsymbol{\beta} + \varepsilon_j. \tag{1}$$

Taking expectations and rearranging:

$$y_j - \mathbb{E}\left(y_j|\mathit{diversity}_j\right) = \left(\mathbf{x}_j - \mathbb{E}\left(\mathbf{x}_j|\mathit{diversity}_j\right)\right)\boldsymbol{\beta} + \varepsilon_j. \tag{2}$$

We estimate the expectations $\mathbb{E}\left(\cdot|\mathit{diversity}_j\right)$ nonparametrically, recover $\widehat{\mathbb{E}}_y(\mathit{diversity}_j)$ and $\widehat{\mathbb{E}}_{\mathbf{x}}(\mathit{diversity}_j)$, and estimate $\widehat{\boldsymbol{\beta}}$, which allows us to then recover $\widehat{m(\cdot)}$:

$$y_j - \mathbf{x}_j'\widehat{\boldsymbol{\beta}} = \alpha + m(\mathit{diversity}_j) + \varepsilon_j$$

▸ back

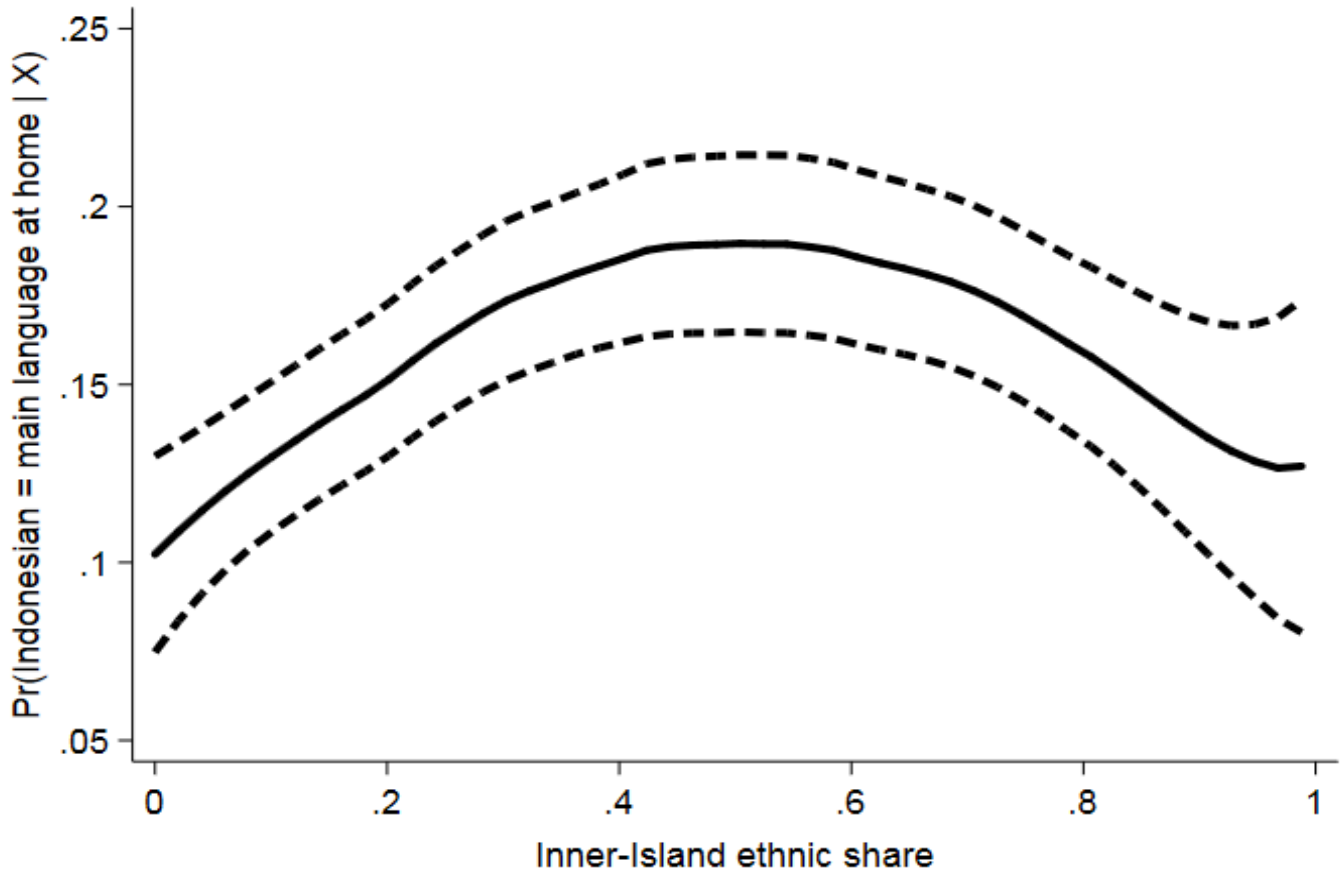
Ethnic Diversity and National Language Use at Home

All Household Members in 1995 SUPAS

back

$$\mathbb{P}(\textit{Indonesian}_{ij}^{1995} = 1) = \alpha + g(\textit{Inner Island ethnic share}_{j}^{\widehat{1995}}) + \mathbf{x}'_{ij}\boldsymbol{\beta} + \varepsilon_{ij}$$

Full Population



Notes: Robinson (1988) partially linear approach to estimating $g(\cdot)$. *Inner Island ethnic share*$^{\widehat{1995}}$ based on those individuals living in the village in 1995 excluding those that we do not observe who departed between 1995 and 2000.

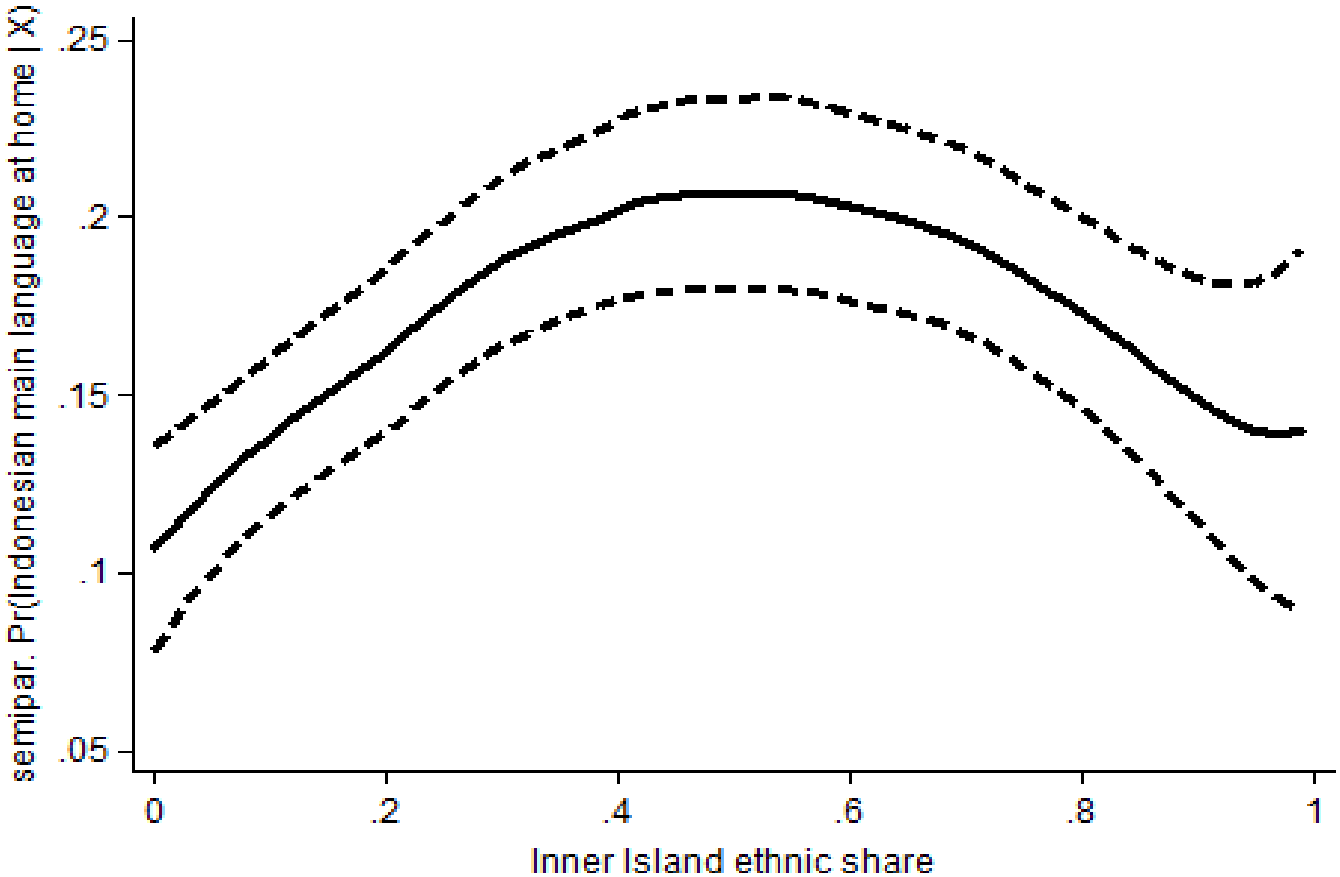
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back

$$\mathbb{P}(\textit{Indonesian}_{ij}^{1995} = 1) = \alpha + g(\textit{Inner Island ethnic share}_{j}^{\widehat{1995}}) + \mathbf{x}_{ij}'\boldsymbol{\beta} + \varepsilon_{ij}$$

Young Population Less Than 30 Years Old



Notes: Robinson (1988) partially linear approach to estimating $g(\cdot)$. *Inner Island ethnic share*$^{\widehat{1995}}$ based on those individuals living in the village in 1995 excluding those that we do not observe who departed between 1995 and 2000.

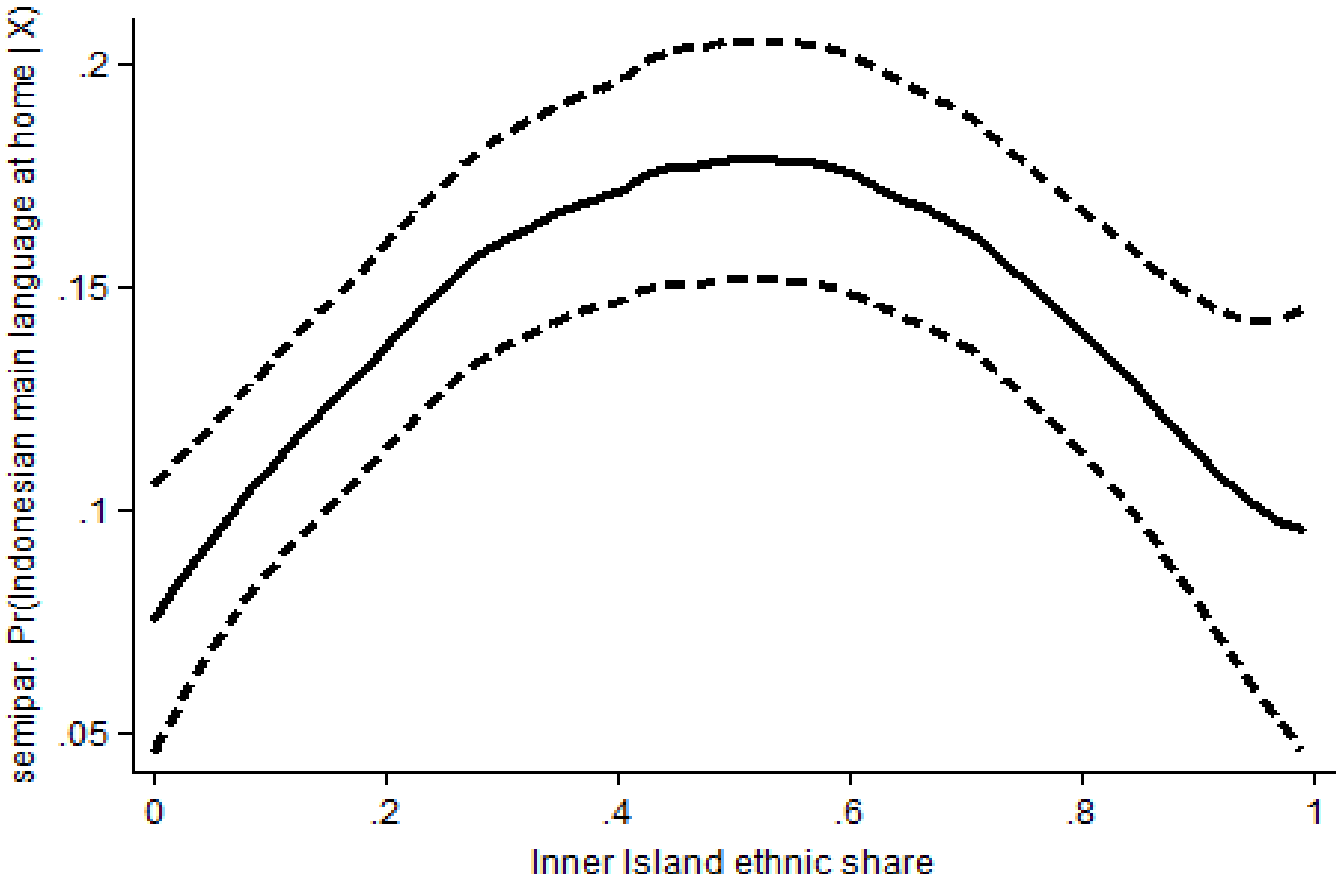
Ethnic Diversity and National Language Use at Home

All Household Members in 1995 SUPAS

▸ back

$$\mathbb{P}(\textit{Indonesian}_{ij}^{1995} = 1) = \alpha + g(\textit{Inner Island ethnic share}_{j}^{\widehat{1995}}) + \mathbf{x}_{ij}'\boldsymbol{\beta} + \varepsilon_{ij}$$

Excluding Immigrants Arriving $\geq$ 3 Years of Initial Settlement



▸ *Notes:* Robinson (1988) partially linear approach to estimating $g(\cdot)$. *Inner Island ethnic share*$^{\widehat{1995}}$ based on those individuals living in the village in 1995 excluding those that we do not observe who departed between 1995 and 2000.

Ethnic Diversity and National Language Use at Home, Auxiliary *Supas* Data

	Dependent Variable: Indonesian is Main Language at Home				
	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Sample Restrictions	None	Excl. Faraway APDDT Immigrants	Initial Period Immigrants	HH Head Not Intermarried	HH Heads Not Indonesian Mother Tongue
Inner-Island ethnic share	1.519	1.328	1.331	1.586	0.659
	(0.395)***	(0.394)***	(0.423)***	(0.378)***	(0.192)***
Inner-Island ethnic share squared	-1.523	-1.334	-1.366	-1.594	-0.601
	(0.431)***	(0.430)***	(0.460)***	(0.423)***	(0.222)***
Turning point	0.499	0.498	0.487	0.498	0.548
p-value	[0.001]***	[0.003]***	[0.004]***	[0.001]***	[0.021]**
Number of Individuals	28,532	24,595	17,175	25,509	25,313
Dependent Variable Mean	0.159	0.139	0.130	0.126	0.081
Sanderson-Windmeijer Linear p-value	0.073	0.067	0.076	0.068	0.067
Sanderson-Windmeijer Quadratic p-value	0.091	0.088	0.041	0.110	0.064
Anderson-Rubin Weak IV Robust p-value	0.004	0.004	0.020	0.000	0.005

Notes: Standard errors in all columns are clustered by district, of which there are 67. */**/*** denotes significant at the 10/5/1 percent significance levels.

▸ back

Ethnic Diversity and National Language Use by Education

	Baseline		Sample Splitting by Education Level			
			≤ Primary Education		> Primary Education	
	OLS	IV-GMM	OLS	IV-GMM	OLS	IV-GMM
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**
Inner-Island ethnic share	0.665	0.845	0.549	0.891	0.813	1.344
	(0.284)**	(0.379)**	(0.289)*	(0.326)***	(0.338)**	(0.534)**
Inner-Island ethnic share squared	-0.854	-0.927	-0.775	-1.041	-0.868	-1.283
	(0.312)***	(0.382)**	(0.311)**	(0.329)***	(0.394)**	(0.589)**
Turning point	0.390	0.456	0.354	0.428	0.468	0.524
p-value	[0.012]**	[0.015]**	[0.032]**	[0.004]***	[0.028]**	[0.037]**
Number of Individuals	2,126	2,126	1,229	1,229	897	897
Dependent Variable Mean	0.249	0.249	0.163	0.163	0.368	0.368
Sanderson-Windmeijer Linear p-value		0.000		0.000		0.000
Sanderson-Windmeijer Quadratic p-value		0.000		0.000		0.000
Anderson-Rubin Weak IV Robust p-value		0.000		0.000		0.000
Hansen J test p-val		0.161		0.139		0.243

Notes: Standard errors in all columns are clustered by district, of which there are 50. */**/*** denotes significant at the 10/5/1 percent significance levels.

▸ back

Robustness Checks on Ethnic Diversity and National Language Use At Home

	Baseline x Controls				Excluding		Dropping Villages w/	
	Baseline Spec.	+ Occupation FE	+ Ethnic Homeland FE	Village Controls	Individual Controls	All x Controls	Squiggly Borders	High Recent Immigration
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**
Inner-Island ethnic share	0.845	0.765	1.103	1.166	0.829	1.257	1.396	0.862
	(0.379)**	(0.373)**	(0.202)***	(0.542)**	(0.350)**	(0.554)**	(0.635)**	(0.369)**
Inner-Island ethnic share squared	-0.927	-0.815	-1.409	-1.417	-0.927	-1.590	-1.386	-0.958
	(0.382)**	(0.384)**	(0.268)***	(0.503)***	(0.353)**	(0.494)***	(0.611)**	(0.407)**
Turning point	0.456	0.391	0.469	0.412	0.447	0.395	0.504	0.450
p-value	[0.015]**	[0.000]***	[0.024]**	[0.018]**	[0.011]**	[0.014]**	[0.017]**	[0.016]**
Number of Individuals	2,126	2,078	2,126	2,126	2,126	2,126	1,823	1,902
Sanderson-Windmeijer Linear p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Sanderson-Windmeijer Quadratic p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Hansen J test p-value	0.161	0.279	0.167	0.130	0.132	0.111	0.344	0.134

Notes: Standard errors in all columns are clustered by district, of which there are 50. */**/*** denotes significant at the 10/5/1 percent significance levels.

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Alternative Diversity Metrics and National Language Use At Home

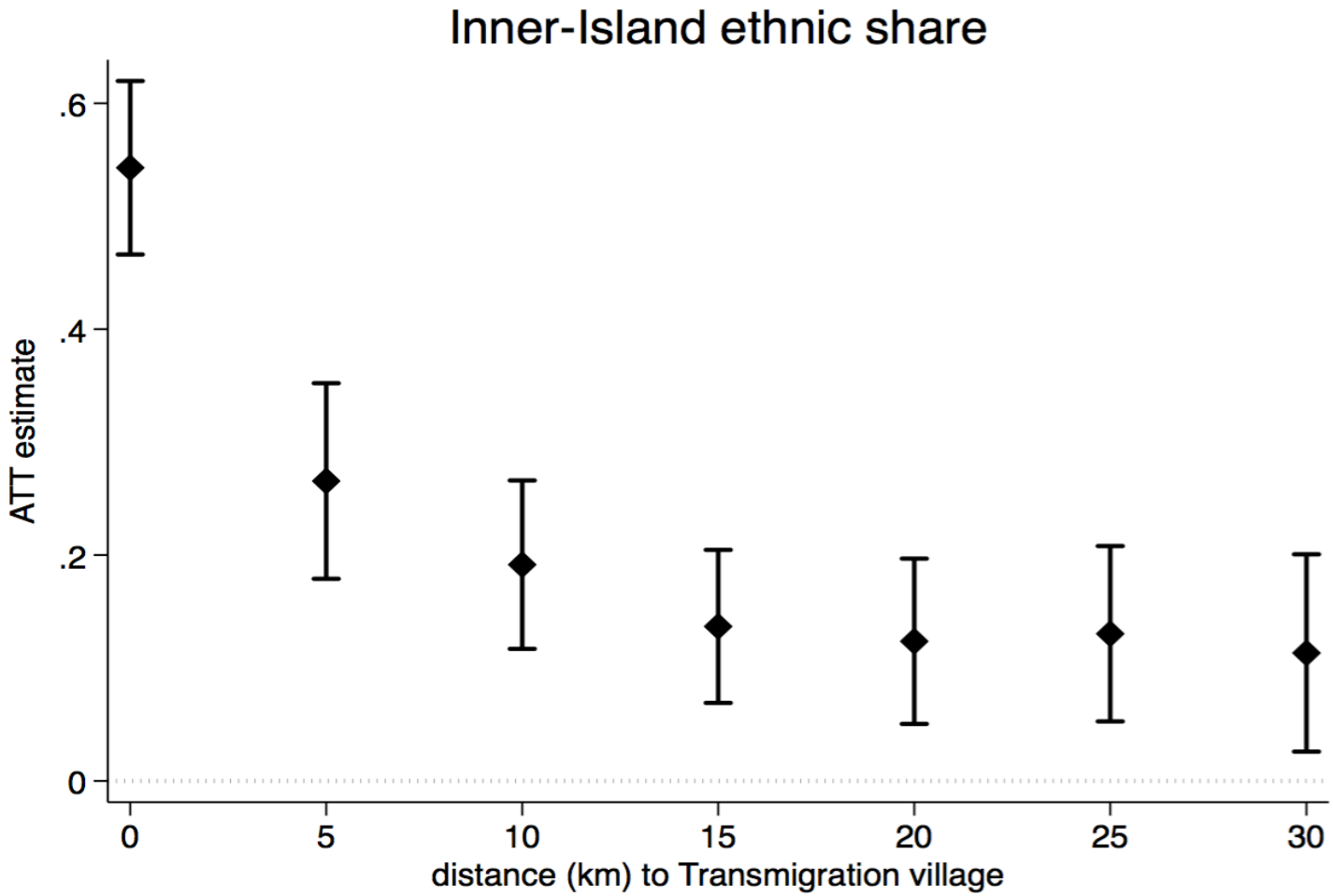
	Dependent Variable: Indonesian is Main Language at Home						
	Baseline		Alternative Diversity Specifications				
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**
Inner-Island ethnic share	0.665						
	(0.284)**						
Inner-Island ethnic share squared	-0.854						
	(0.312)***						
Inner-Island ethnic share, bottom tercile		0.066					
		(0.058)					
Inner-Island ethnic share, middle tercile		0.203					
		(0.059)***					
ethnic fractionalization (Inner, Outer)			0.318				
			(0.142)**				
overall ethnic fractionalization				0.462			
				(0.105)***			
ethnic fractionalization, all Inner-Island					0.165		
					(0.122)		
Javanese share of Inner-Island ethnics						0.777	
						(0.395)*	
Javanese share of Inner-Island ethnics squared						-0.638	
						(0.312)**	
number of initial transmigrants, bottom tercile							0.065
							(0.055)
number of initial transmigrants, middle tercile							0.168
							(0.060)***
Number of Individuals	2,126	2,126	2,126	2,126	2,047	2,047	2,126

Notes: Standard errors in all columns are clustered by district, of which there are 50. */**/*** denotes significant at the 10/5/1 percent significance levels.

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Spillovers to Neighboring Villages

5 km Discs Outside Transmigration Villages

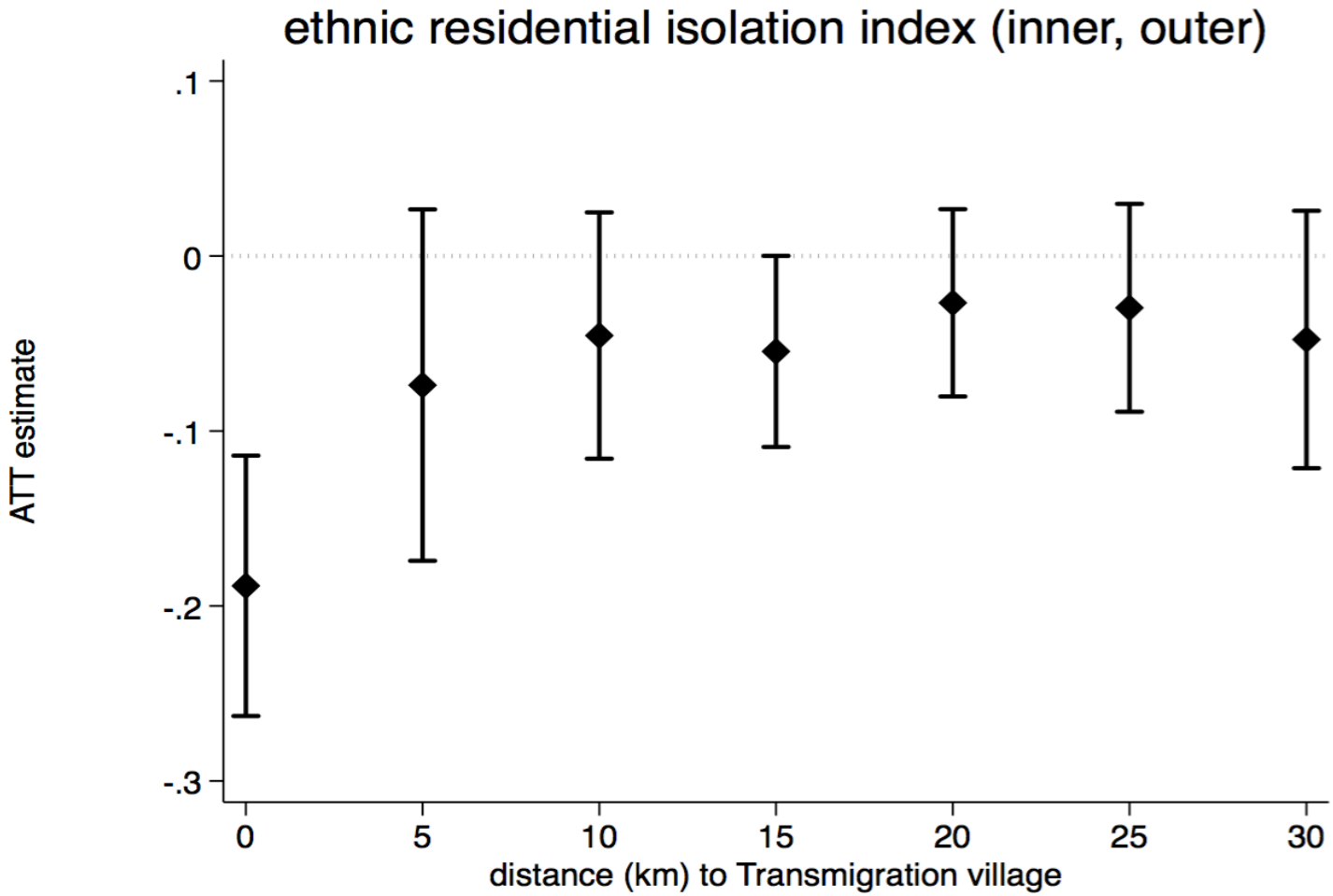


Notes: ATT estimates based on usual estimation procedure on villages 0, 1-5, 6-10, . . . , km from Transmigration village boundaries.

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Spillovers to Neighboring Villages

5 km Discs Outside Transmigration Villages

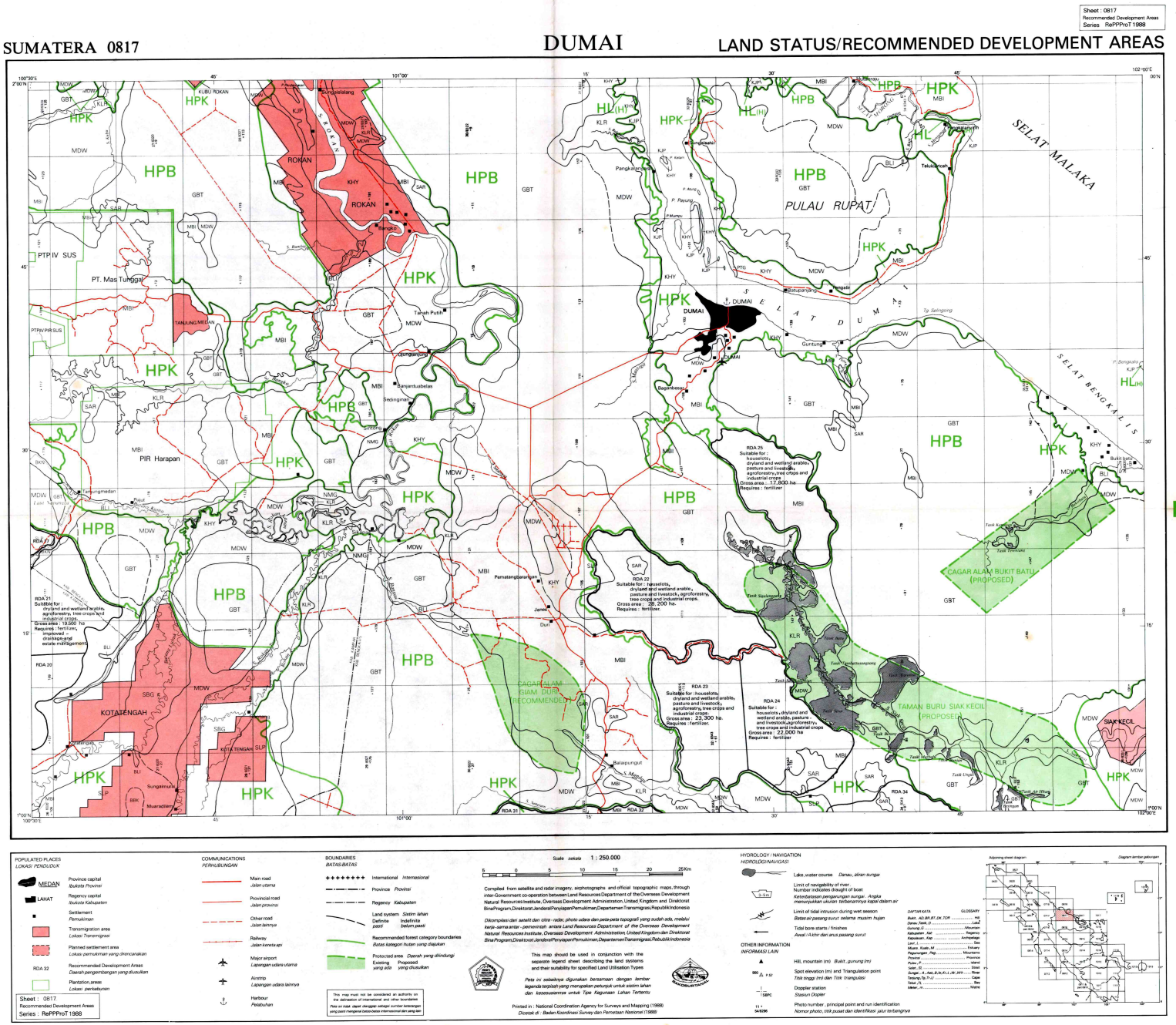


Notes: ATT estimates based on usual estimation procedure on villages 0, 1-5, 6-10, . . . , km from Transmigration village boundaries.

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Example of Recommended Development Areas

back



Reweighting Control Villages $\Longrightarrow$ Balance back

Predtermined Village Characteristics by Treatment and Control Status

	Transmigration Treated (T)	Non-Treated (NT)	RDA Control (C)	(t-stat) Statistical Difference		
					Raw	Adjusted
	mean (sd)	mean (sd)	mean (sd)	$\mu_T - \mu_{NT}$	$\mu_T - \mu_C$	$\mu_T - \mu_C$
log village area, Ha	7.53 (1.0)	6.99 (1.5)	8.24 (1.3)	5.32***	-4.13***	-0.21
% w/ slope between 0-2%	37.63 (28.3)	30.19 (30.6)	16.96 (17.7)	2.69***	5.88***	0.92
% w/ slope between 2-8%	48.25 (25.4)	39.27 (24.7)	48.21 (21.7)	3.89***	0.01	-0.12
% w/ slope between 8-30%	9.87 (16.5)	20.32 (22.6)	24.73 (19.4)	-6.35***	-5.35***	0.56
Vector Ruggedness Measure	0.31 (0.1)	0.28 (0.2)	0.31 (0.1)	2.81***	0.21	-0.58
log altitude, m^2	3.29 (2.9)	3.77 (2.7)	4.83 (2.2)	-1.91*	-5.08***	1.54
Organic Carbon (%)	4.77 (6.2)	3.53 (4.7)	3.06 (5.2)	2.69***	2.77***	-0.33
Topsoil Sodicity (ESP) %	1.57 (0.4)	1.50 (0.5)	1.63 (0.5)	1.72*	-0.86	1.81*
Topsoil pH (-log(H+))	5.07 (0.4)	5.33 (0.7)	5.35 (0.6)	-5.22***	-2.26**	0.56
Coarse texture soils (%)	0.10 (0.2)	0.06 (0.2)	0.09 (0.2)	3.44***	0.30	-1.12
Medium texture soils (%)	0.70 (0.2)	0.71 (0.2)	0.65 (0.2)	-0.49	1.98*	-0.85
Very poor or poor drainage (%)	0.39 (0.4)	0.30 (0.3)	0.20 (0.3)	3.15***	5.50***	0.87
Imperfect drainage soils (%)	0.06 (0.2)	0.12 (0.3)	0.21 (0.3)	-2.73***	-3.24***	-0.86
Avg. rainfall, 1948-1978	225.26 (35.1)	215.29 (41.4)	237.66 (35.8)	2.39**	-1.70*	0.31
Avg. temp (Celcius), 1948-1978	26.26 (1.7)	25.36 (2.7)	25.75 (1.8)	4.74***	1.77*	0.15
Minimum Log Distance to Villages on Java or Bali	6.69 (0.5)	6.91 (0.6)	6.91 (0.3)	-2.66***	-1.97*	-0.73
Log Distance to Nearest Major Road	0.08 (0.1)	0.07 (0.1)	0.10 (0.1)	1.48	-1.09	-0.29
Log Distance to Nearest Coast	10.56 (1.1)	9.96 (1.5)	10.84 (0.9)	4.32***	-1.65	1.07
Log Distance to Nearest River	8.09 (0.8)	7.95 (1.1)	8.22 (0.8)	2.06**	-1.54	0.45
Log Distance to Subdistrict Capital	2.43 (1.5)	1.73 (1.6)	1.97 (1.8)	7.54***	3.35***	1.18
Log Distance to District Capital	4.12 (1.0)	3.46 (1.4)	4.10 (1.1)	7.33***	0.13	2.40**
linguistic similarity with Java/Bali languages	0.58 (0.1)	0.60 (0.1)	0.60 (0.1)	-2.36**	-1.61	-1.09

Notes: This table reports the sample means (standard deviations, sd) for the time-invariant or predetermined village-level characteristics that comprise our main covariate vector **x**, which determined site selection. We consider three groups of villages: Transmigration villages settled in the period 1979–1988 or treated sites (T), non-Transmigration villages or non-treated sites (NT), and Recommended Development Areas (RDA) or control sites (C) that were suggested as resettlement areas but never received the program due to sudden budgetary cutbacks. The *t*-statistics reported in the latter three columns are recovered from the coefficient on the treatment variable in a regression of the given characteristic on the treatment indicator (and, in the

Why the Control Villages are Useful

Further Clarifying the Counterfactual back

- In most settings, longstanding diverse communities look special:
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  - strong sorting of immigrants based on location-specific skills table
  - more urban, higher education and greater interethnic marriage table

- Transmigration and Control villages help us address this endogeneity
  - **Transmigration**: state-sponsored settlement of **new villages**
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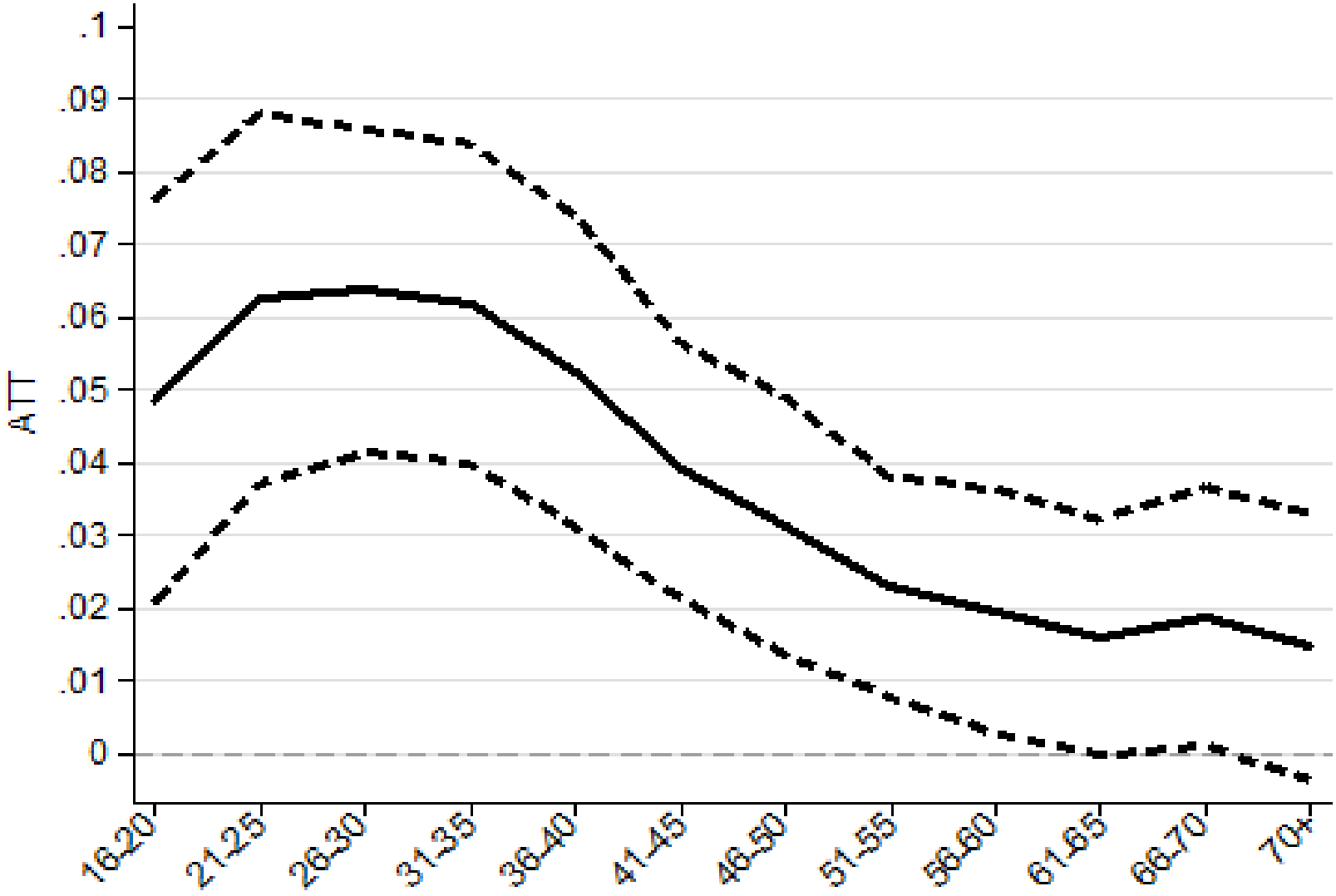
- Balanced natural advantages $\Longrightarrow$ similar counterfactual demographics and institutions if not for the Transmigration program

- After 1979 Law, all villages endowed w/ same *de jure* institutions

  $\Longrightarrow$ helps rule out initial, first order institutional differences

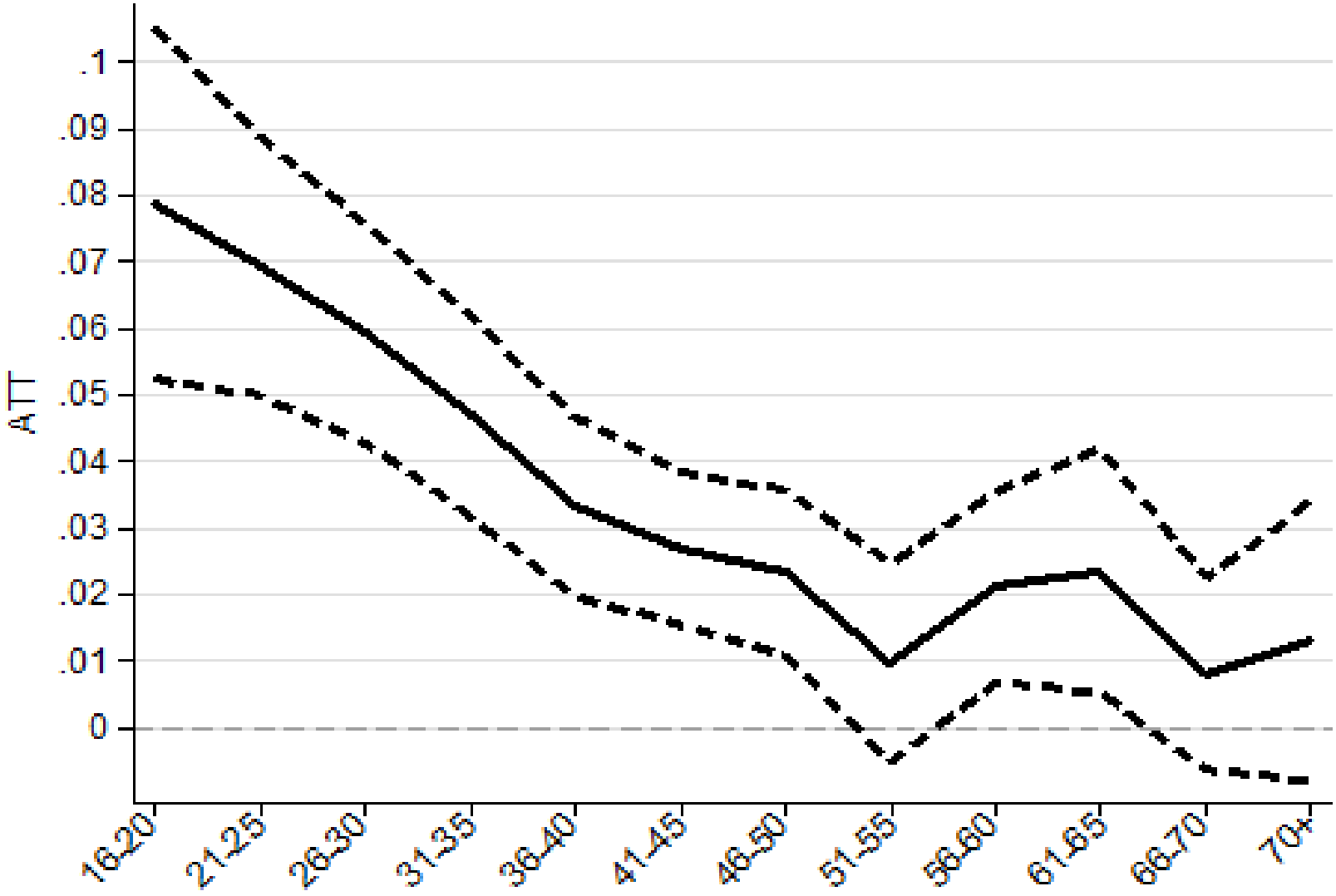
Effects Are Similar for **Men** and Women

Restricting to Outer Islands Ethnicities Born in Outer Islands District of Transmigration or Control Village back



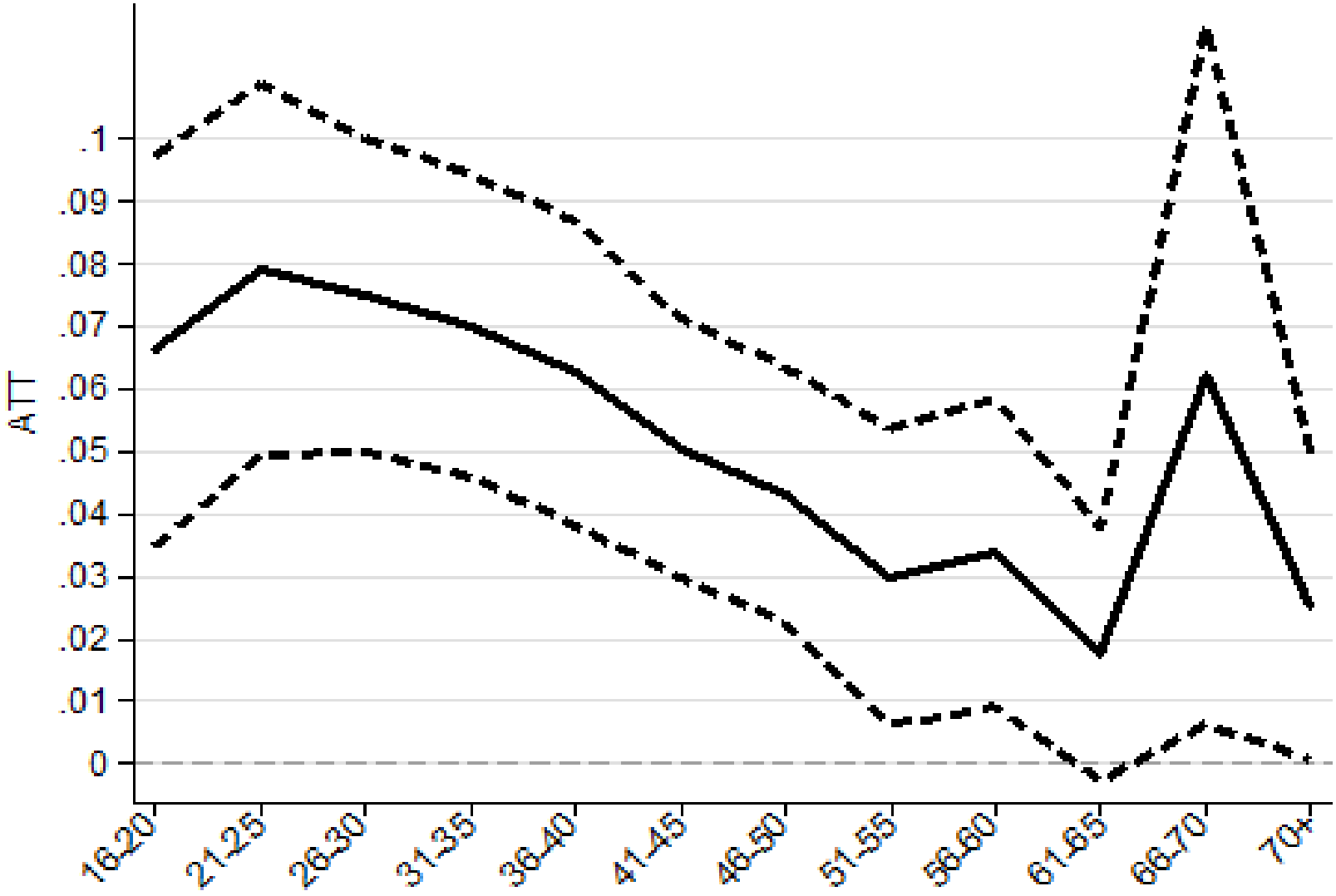
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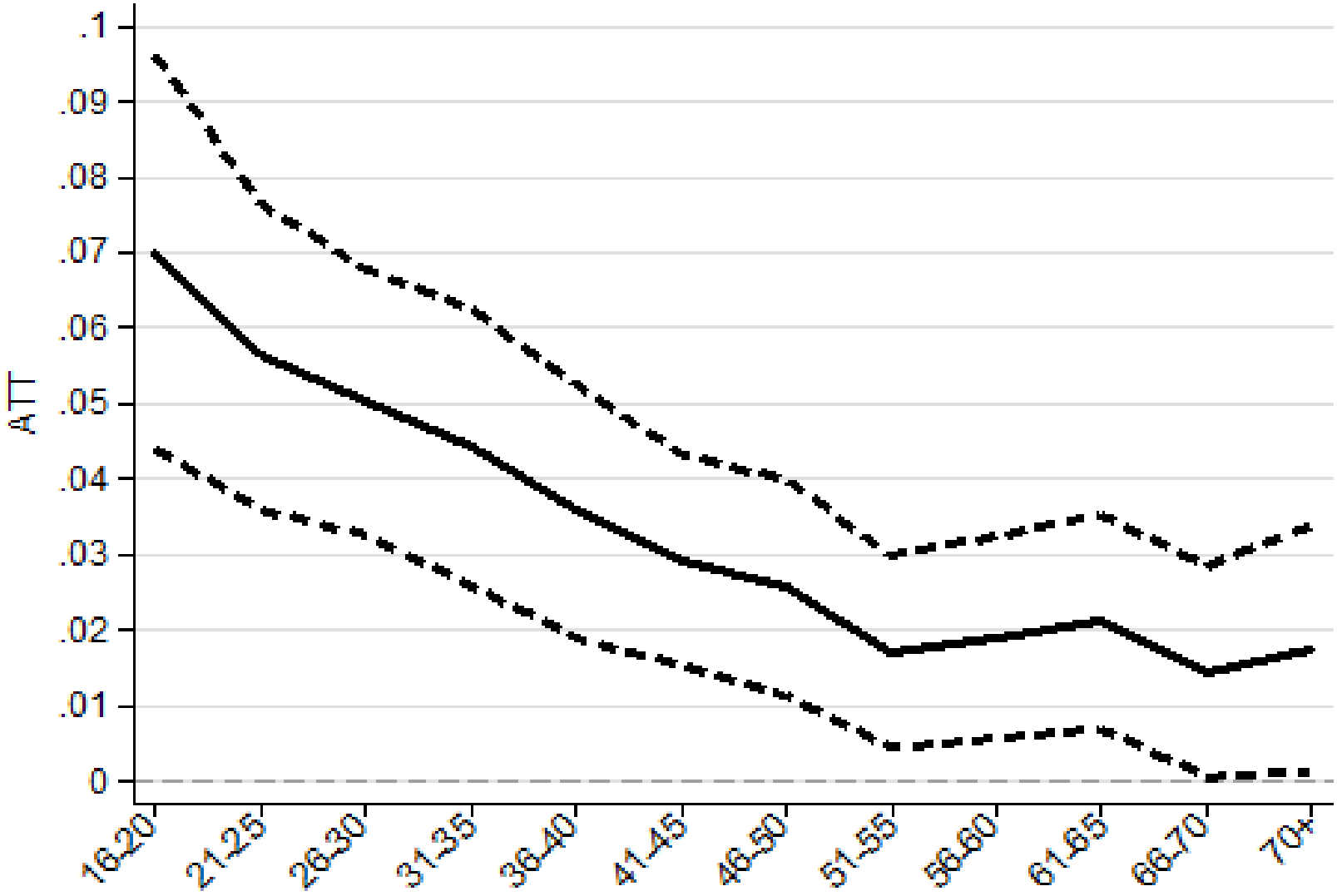
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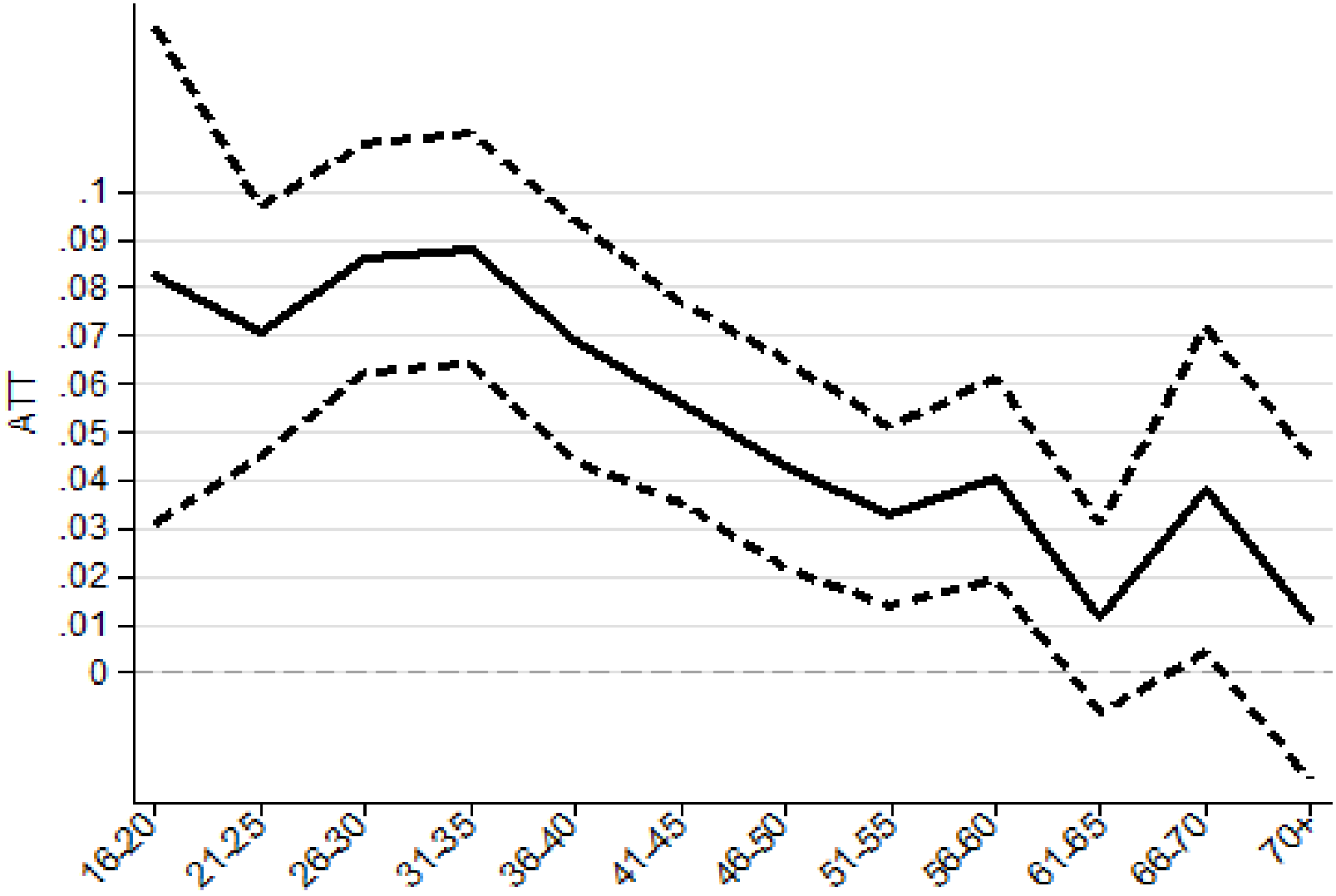
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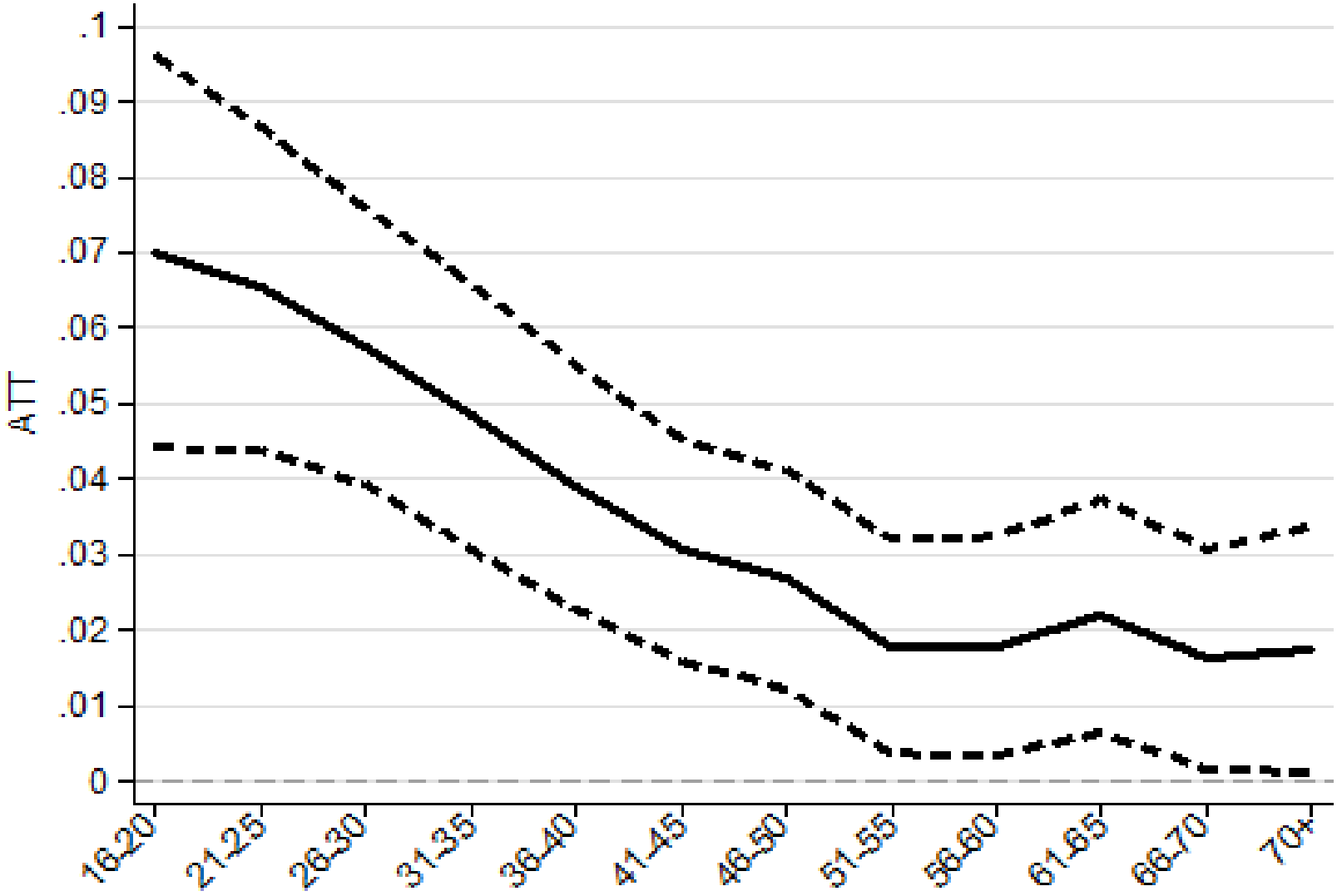
Effects Are Similar for **Traders/Service** Occupations

Restricting to Outer Islands Ethnicities Born in Outer Islands District of Transmigration or Control Village back



Effects Are Similar for **Non-Traders/Service** Occupations

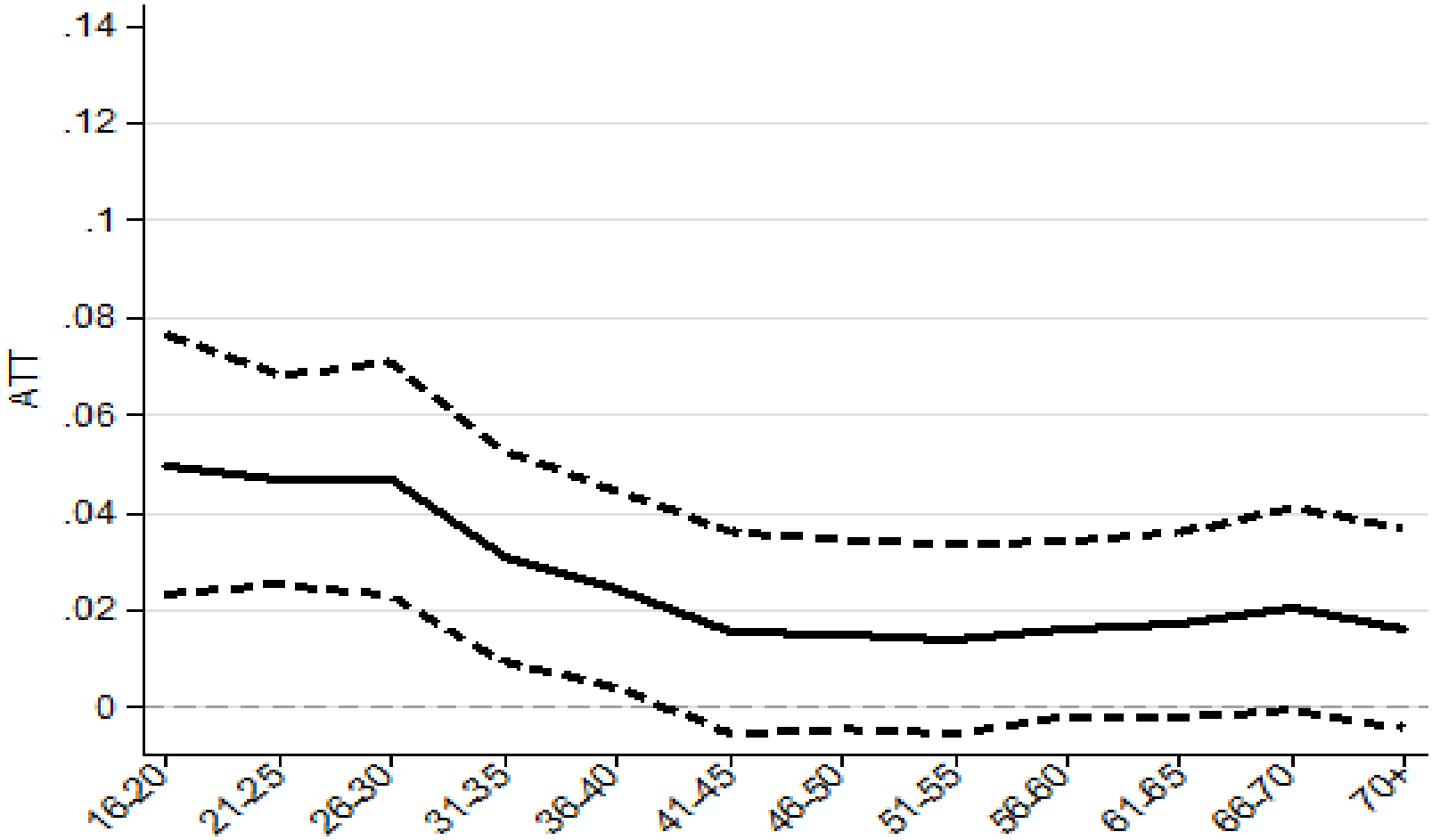
Restricting to Outer Islands Ethnicities Born in Outer Islands District of Transmigration or Control Village back



Age-Specific ATT for Any Intermarriage ∼650,000 Marriages

Baseline

back

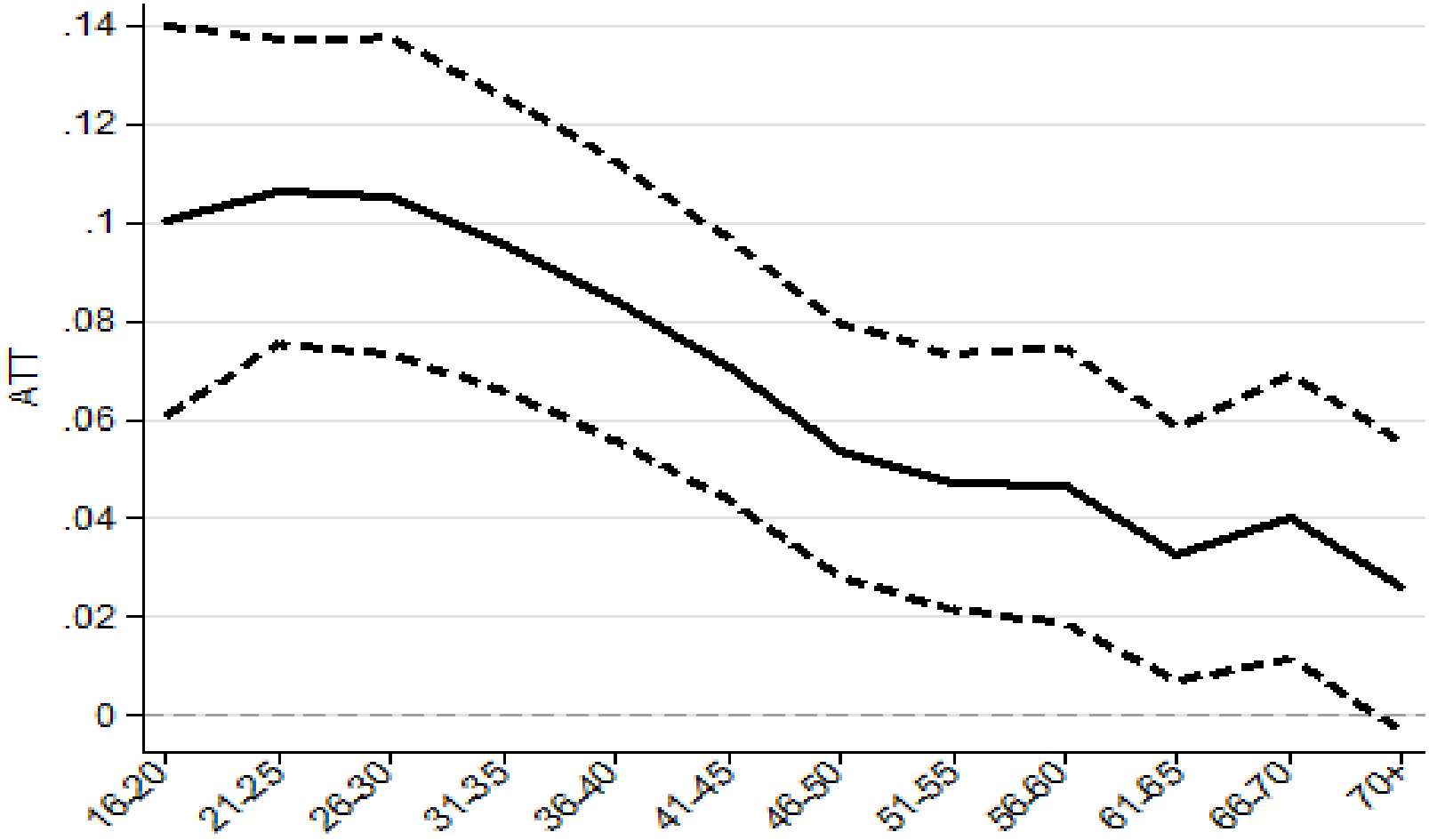


Notes: 95% confidence interval on age group-specific ATT estimates.

Restricting to Native-Born Outer Islands Ethnics

+ Birth District Fixed Effects

back

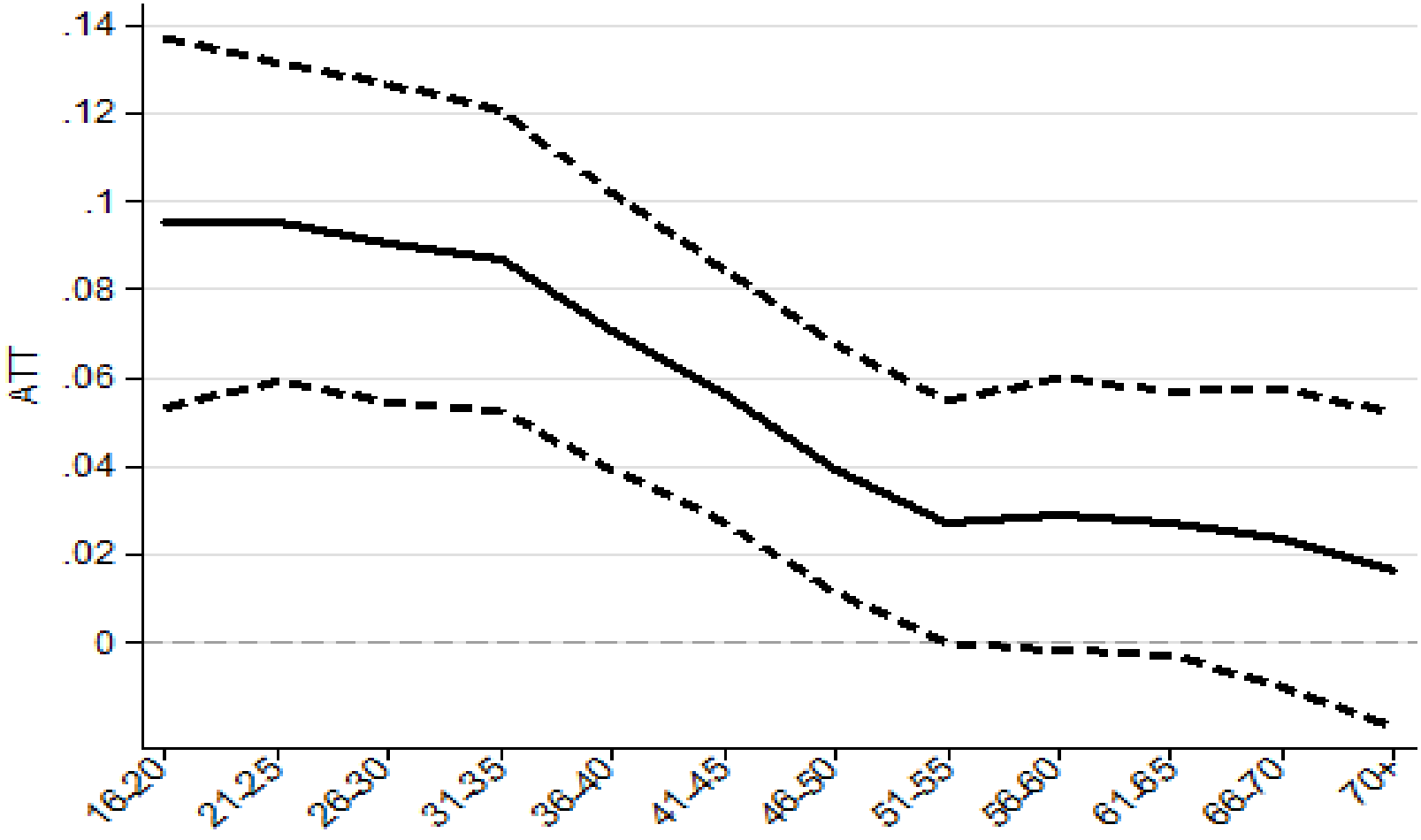


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. . . and Adding Ethnicity Fixed Effects

+ Birth District Fixed Effects

back



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Decomposing Aggregate Intermarriage Effects

- ATT $\Longrightarrow$ $\uparrow$ young interethnic marriage (*IEM*) from 2.3% to 7.3%

  How much of the increase is due to the pure ***supply effect****?*

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1. Re-estimating ATT conditional on (polynomial of) *IEM*(*random*)
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- Reduced form evidence $\Longrightarrow$ Δ demand for *IEM* (inner, outer)
  (Δ demand explains $\approx$2/3rd, Δ supply explains $\approx$1/3rd)

back

Transmigration $\Longrightarrow$ National Language Use at Home
Individual-Level ATT Estimates

back

	$\mathbb{P}$(Daily ... Language Use at Home)		
	Indonesian	Inner Island	Outer Island
	(1)	**(2)**	**(3)**
1. Baseline ATT	0.250	-0.002	-0.248
	(0.126)**	(0.068)	(0.162)
2. Conditional on age, gender, education	0.256	-0.005	-0.252
	(0.118)**	(0.070)	(0.159)
3. Individuals aged $\leq$ 35	0.251	-0.069	-0.182
	(0.137)*	(0.080)	(0.177)
4. Conditional on age, gender, education, occupation	0.245	-0.002	-0.243
	(0.127)*	(0.068)	(0.162)
5. Conditional on Malay indigenous language	0.248	-0.003	-0.244
	(0.131)*	(0.070)	(0.165)
6. Conditional on indigenous language distance to Malay	0.243	-0.011	-0.231
	(0.123)**	(0.071)	(0.158)